

Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline

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abstract

PURPOSE To provide recommendations for the management of patients with metastatic clear cell renal cell carcinoma (ccRCC).

METHODS An Expert Panel conducted a systematic literature review to obtain evidence to guide treatment recommendations.

RESULTS The panel considered peer-reviewed reports published in English.

RECOMMENDATIONS The diagnosis of metastatic ccRCC should be made using tissue biopsy of the primary tumor or a metastatic site with the inclusion of markers and/or stains to support the diagnosis. The International Metastatic RCC Database Consortium risk criteria should be used to inform treatment. Cytoreductive nephrectomy may be offered to select patients with kidney-in-place and favorable- or intermediate-risk disease. For those who have already had a nephrectomy, an initial period of active surveillance may be offered if they are asymptomatic with a low burden of disease. Patients with favorable-risk disease who need systemic therapy may be offered an immune checkpoint inhibitor (ICI) in combination with a vascular endothelial growth factor receptor (VEGFR) tyrosine kinase inhibitor (TKI); patients with intermediate or poor risk should be offered a doublet regimen (no recommendation was provided between ICIs or an ICI in combination with a VEGFR TKI). For select patients, monotherapy with either an ICI or a VEGFR TKI may be offered on the basis of comorbidities. Interleukin-2 remains an option, although selection criteria could not be identified. Recommendations are also provided for second- and subsequent-line therapy as well as the treatment of bone metastases, brain metastases, or the presence of sarcomatoid features. Participation in clinical trials is highly encouraged for patients with metastatic ccRCC.

Additional information is available at www.asco.org/genitourinary-cancer-guidelines

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INTRODUCTION

Kidney cancer will be diagnosed in 79,000 US patients in 2022 (50,290 men and 28,710 women) and will account for 13,920 deaths (8,960 men and 4,960 women).¹ Clear cell renal cell carcinoma (ccRCC) is the most common subtype of RCC, and it continues to be a major source of morbidity and mortality.^{1,2} This disease has been a bellwether tumor type, with novel classes of therapeutics often first explored in ccRCC, tracking closely with discoveries related to the underlying biology. An acceleration in reported phase III studies has created a seismic shift in the available treatment options for treating this disease. Opportunities for extended survival and even durable disease control in the metastatic setting are reshaping the trajectory of this cancer and raising the stakes in providing optimal treatment

planning. This rapid evolution has paved the way for the first creation of a comprehensive set of guidelines to guide therapeutic selection and future translational development to continue to advance the treatment of metastatic ccRCC.

GUIDELINE QUESTIONS

This clinical practice guideline addresses six overarching clinical questions: (1) How is metastatic clear cell renal cell carcinoma defined and how is it diagnosed? (2) What is the role of cytoreductive nephrectomy in metastatic clear cell renal cell carcinoma? (3) What are the preferred options for first-line systemic treatment of metastatic clear cell renal cell carcinoma? (4) What is the optimal second- or later-line systemic treatment for metastatic clear cell renal cell carcinoma? (5) What is the optimal

ASSOCIATED

CONTENT

Appendix

Data Supplement

Author affiliations and support information (if applicable) appear at the end of this article.

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THE BOTTOM LINE

Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline

Guideline Question

What are the optimal treatments of patients with metastatic clear cell renal cell carcinoma (ccRCC)?

Target Population

Patients with metastatic ccRCC.

Target Audience

Medical oncologists, radiation oncologists, urologists, nurses, other health care practitioners, social workers, patients, and caregivers.

Methods

An Expert Panel was convened to develop clinical practice guideline recommendations on the basis of a systematic review of the medical literature.

Clinical Question 1

How is metastatic clear cell renal cell carcinoma defined and how is it diagnosed?

Recommendation 1.1. The diagnosis of metastatic ccRCC should ideally involve comparison of tissue acquired outside the site of primary disease to the primary histology. Histologic evaluation should include common markers of ccRCC including paired box gene 8 and carbonic anhydrase IX (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 1.2. Radiographic diagnosis of metastatic ccRCC may be applied in selected circumstances, such as settings in which a prior diagnosis of renal cell carcinoma has been established, when metastatic tissue is not readily accessible by biopsy, or when RECIST 1.1 measurable disease is evident, especially within a year of the initial diagnosis (Type: Consensus based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Weak).

Clinical Question 2

What is the role of cytoreductive nephrectomy in metastatic clear cell renal cell carcinoma?

Recommendation 2.1. Select patients (see *Practical Information*) with metastatic ccRCC may be offered cytoreductive nephrectomy (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong). Practical Information: Select patients include those with optimally one International Metastatic RCC Database Consortium (IMDC) risk factor who can have a significant majority of their tumor burden removed at the time of surgery. First-line treatment is shown in [Figure 1](#).

Clinical Question 3

What are the preferred options for first-line systemic treatment of metastatic clear cell renal cell carcinoma?

Recommendation 3.1. Select patients with metastatic ccRCC (see *Practical Information*) may be offered an initial active surveillance strategy (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Practical Information: Select patients include those with IMDC favorable and intermediate risk, patients with limited or no symptoms related to disease, a favorable histologic profile, a long interval between nephrectomy and the development of metastasis, or with limited burden of metastatic disease.

Recommendation 3.2. All patients with metastatic ccRCC who require systemic therapy in the first-line setting should undergo risk stratification into IMDC favorable (0), intermediate (1-2), and poor (3+) risk groups.³ Patients with intermediate- or poor-risk disease should be offered combination treatment with two immune checkpoint inhibitors (ICIs; ie, ipilimumab and nivolumab) or an ICI in combination with a vascular endothelial growth factor receptor tyrosine kinase inhibitor (VEGFR TKI; Type: Evidence based; benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 3.3. Patients with favorable risk disease who require systemic therapy may be offered an ICI in combination with a VEGFR TKI (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 3.4. Select patients with metastatic ccRCC receiving systemic therapy in the first-line setting including those with favorable-risk disease or with certain coexisting medical problems may be offered monotherapy with either a VEGFR TKI or an ICI (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

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THE BOTTOM LINE (CONTINUED)

Recommendation 3.5. The use of high-dose interleukin-2 (HD-IL2) may be considered in the first-line systemic therapy setting for patients with metastatic ccRCC (see *Practical Information*). Attempts to develop criteria to predict those patients most likely to derive benefit from HD-IL2 have been unsuccessful (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Weak).

Practical Information: The significant toxicity of this regimen must be weighed in relation to the newer immunotherapy regimens that have largely replaced this treatment. The Expert Panel was not able to identify a patient population who should receive this treatment preferentially on the basis of available data. The Expert Panel did agree that HD-IL-2 should be administered in experienced high-volume centers, and that enrollment in clinical trials was preferred.

First-line treatment is shown in [Figure 1](#).

Clinical Question 4

What is the optimal second- or later-line systemic treatment for metastatic clear cell renal cell carcinoma?

Recommendation 4.1. Nivolumab or cabozantinib should be offered to patients who progressed on a VEGFR TKI alone (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 4.2. Patients progressing on combination immunotherapy (eg, nivolumab and ipilimumab) should be offered a VEGFR TKI (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 4.3. Patients who progress after initial therapy combining VEGFR TKI with an ICI may be offered an alternate VEGFR TKI as a single agent (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 4.4. For patients on immunotherapy who experience limited disease progression (eg, one site of progression), local therapy (radiation, thermal ablation, and excision) may be offered, and immunotherapy may be continued (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Weak).

Second-line or greater treatment is shown in [Figure 2](#).

Clinical Question 5

What is the optimal application of metastasis-directed therapy for metastatic clear cell renal cell carcinoma?

Recommendation 5.1. For patients with low-volume metastatic renal cell carcinoma, definitive metastasis-directed therapies may be offered and include surgical resection (metastasectomy), ablative measures, or radiotherapy (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 5.2. For patients undergoing complete metastasectomy, subsequent TKIs are not routinely recommended (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Metastases-directed treatment is shown in [Figure 3](#).

Clinical Question 6

What considerations should be applied to treatment of special subsets of metastatic clear cell renal cell carcinoma (eg, bone metastases, brain metastases, and sarcomatoid carcinomas)?

Recommendation 6.1.1. Patients with symptomatic bone metastases from metastatic ccRCC should receive bone-directed radiation (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 6.1.2. Patients with bone metastases from metastatic ccRCC should be offered a bone resorption inhibitor (either bisphosphonate or receptor activator of nuclear factor kappa-B ligand inhibitor) when clinical concern for fracture or skeletal-related events is present (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 6.1.3. No recommendation regarding optimal systemic treatment for metastatic ccRCC patients with bone metastasis can be made; however, it is our expert opinion that cabozantinib-containing regimens may be preferred (Type: Consensus based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Moderate).

Recommendation 6.2.1. Patients with brain metastases from metastatic ccRCC should receive brain-directed local therapy with radiation therapy and/or surgery (Type: Consensus based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 6.2.2. No recommendation regarding optimal systemic therapy for patients with metastatic ccRCC and brain metastases can be made (Type: Consensus based, benefits to harms ratio unknown; Evidence quality: NA; Strength of recommendation: Strong).

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THE BOTTOM LINE (CONTINUED)

Recommendation 6.3. Patients with metastatic ccRCC with sarcomatoid features should receive ICI-based combination first-line treatment (ipilimumab plus nivolumab, or alternatively, an ICI plus a TKI; Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong). Special patient subtypes are shown in [Figure 4](#).

Additional Resources

Definitions for the quality of the evidence and strength of recommendation ratings are available in Appendix [Table A1](#) (online only). More information, including a supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/genitourinary-cancer-guidelines. The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.net.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

application of metastasis-directed therapy for metastatic clear cell renal cell carcinoma? (6) What considerations should be applied to treatment of special subsets of metastatic clear cell renal cell carcinoma (eg, bone metastases, brain metastases, and sarcomatoid carcinomas)?

METHODS

Guideline Development Process

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel, which included

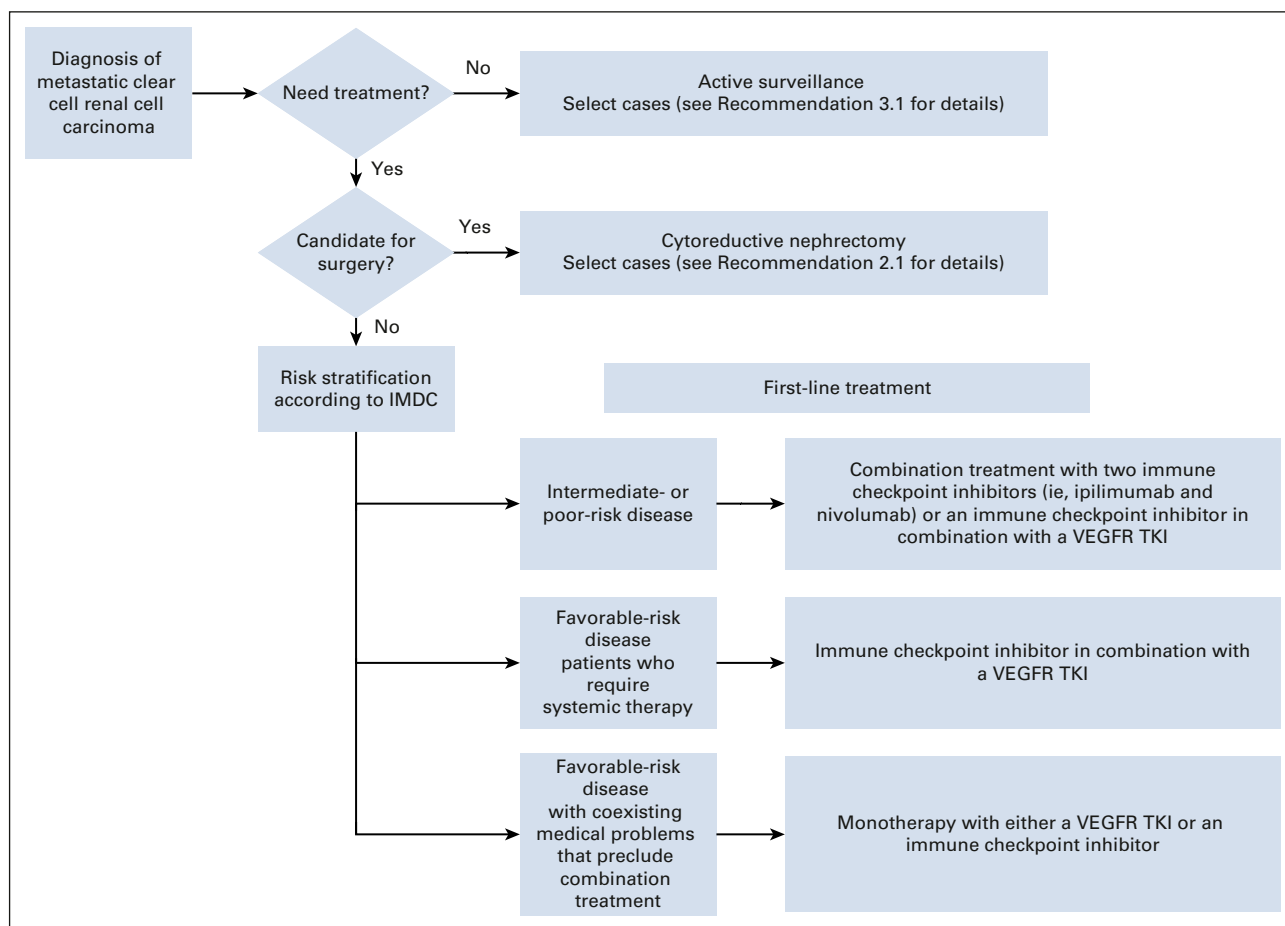


FIG 1. First-line treatment. IMDC, International Metastatic RCC Database Consortium; VEGFR TKI, vascular endothelial growth factor receptor tyrosine kinase inhibitor.

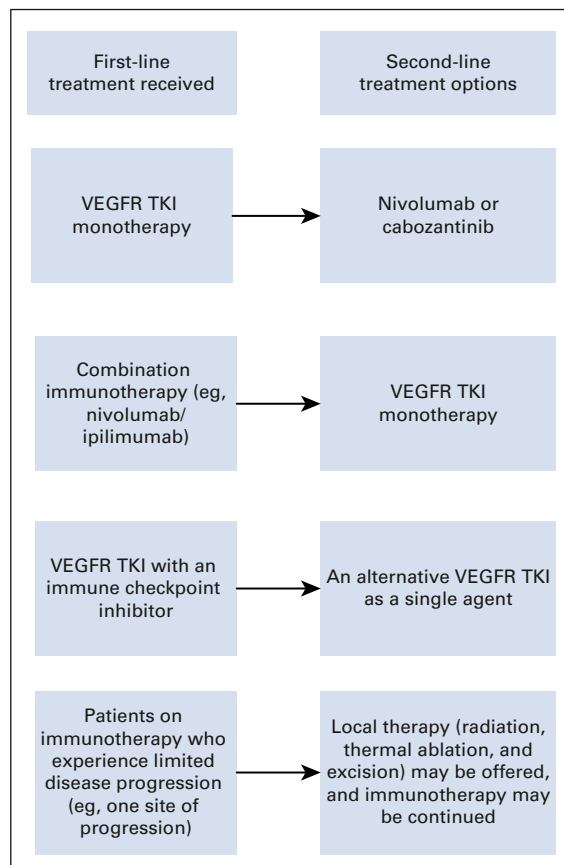


FIG 2. Second-line or greater treatment. VEGFR TKI, vascular endothelial growth factor receptor tyrosine kinase inhibitor.

patient representation, two clinician representatives from the American Society for Therapeutic Radiology and Oncology (ASTRO), and an ASCO guidelines staff member with health research methodology expertise. The Expert Panel met via teleconference and/or webinar and corresponded through e-mail. Based upon the consideration of the evidence, the authors were asked to contribute to the development of the guideline, provide critical review, and finalize the guideline recommendations. The guideline recommendations were sent for an open comment period of two weeks allowing the public to review and comment on the recommendations after submitting a confidentiality

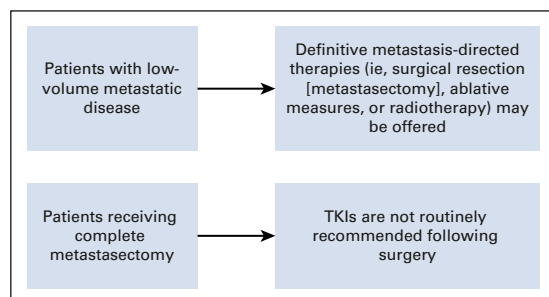


FIG 3. Metastases-directed treatment. TKI, tyrosine kinase inhibitor.

agreement. These comments were taken into consideration while finalizing the recommendations. Members of the Expert Panel were responsible for reviewing and approving the penultimate version of the guideline, which was then circulated for external review, and submitted to the *Journal of Clinical Oncology (JCO)* for editorial review and consideration for publication. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO Evidence Based Medicine Committee before publication. All funding for the administration of the project was provided by ASCO.

The recommendations were developed by using a systematic review (the PUBMED database was searched up to October 15, 2020, and later updated to include relevant papers through the end of March 2022 [see Data Supplement 5, online only: Literature Search Strategy]) of systematic reviews with or without meta-analysis, randomized clinical trials, other relevant study designs, and clinical experience. Articles were selected for inclusion in the systematic review of the evidence on the basis of the following criteria:

- Population: Patients with metastatic ccRCC
- Interventions: cytoreductive nephrectomy (CN), first-line treatment, second-line treatment, treatment focused on metastatic disease, treatment focused on special populations (bone metastases, brain metastases, and sarcomatoid features).
- Comparisons: radiotherapy or radiotherapy plus immunotherapy compared against CN, surgery or Nivo plus Ipi or pembrolizumab plus axitinib or avelumab plus axitinib or atezolizumab plus bevacizumab or nivolumab plus cabozantinib or pazopanib or sunitinib or cabozantinib or temsirolimus or high-dose interleukin-2 (IL-2) or active surveillance compared against a first-line treatment option, cabozantinib versus everolimus or nivolumab versus everolimus or axitinib versus sorafenib or Nivo plus Ipi or lenvatinib plus everolimus compared against themselves or another option for second-line treatment, surgical extirpation, or thermal ablation (radiofrequency, microwave, and cryoablation) or stereotactic ablative radiotherapy for metastases target treatments; for special populations, comparisons were made between any previously listed treatment option where outcome data for patients with bone metastases, brain metastases, or sarcomatoid features were reported separately.
- Outcomes of interest were overall survival (OS), progression-free survival (PFS), quality of life (QoL), and grade 3-5 adverse effects.
- Fully published or recent meeting presentations of English-language reports of rigorously conducted systematic reviews with or without meta-analysis, meta-analyses alone, randomized clinical trials, or other relevant study designs.

Articles were excluded from the systematic review if they were (1) meeting abstracts not subsequently published in

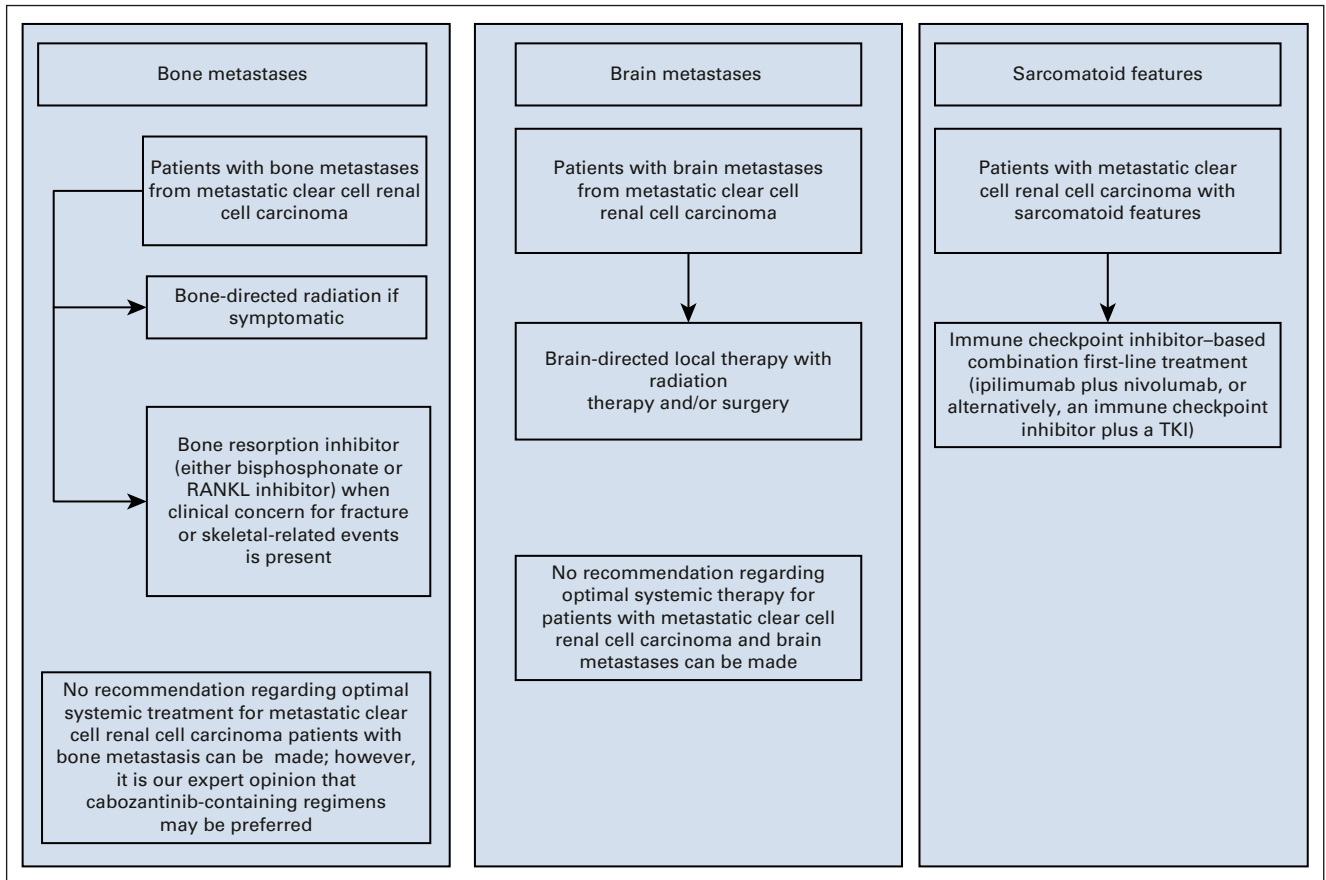


FIG 4. Special patient subtypes. RANKL, receptor activator of nuclear factor kappa-B ligand; TKI, tyrosine kinase inhibitor.

peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, and narrative reviews; and (3) published in a non-English language. The guideline recommendations are crafted, in part, using the *Guidelines Into Decision Support* methodology and accompanying BRIDGE-Wiz software.⁴ In addition, a guideline implementability review is conducted. On the basis of the implementability review, revisions were made to the draft to clarify recommended actions for clinical practice. Ratings for the type and strength of recommendation, evidence, and potential bias are provided with each recommendation.

The ASCO Expert Panel and guidelines staff will work with co-chairs to keep abreast of any substantive updates to the guideline. On the basis of formal review of the emerging literature, ASCO will determine the need to update. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the guideline update process. This is the most recent information as of the publication date.

Guideline Disclaimer

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Guideline and Conflicts of Interest

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines (“Policy,” found at <https://www.asco.org/guideline-methodology>). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

RESULTS

Fifty-four papers^{5-25,26-45,46-59} comprising one clinical practice guideline,⁵ three systematic reviews,⁶⁻⁸ 46 randomized trials^{9-28,30-45,46-59} and one retrospective review²⁹ were retained and form the body of evidence for this guideline. Three papers, comprising one clinical practice guideline⁵ and two trials,^{9,10} reported on CN compared with other treatments; 20 papers, comprising two systematic reviews^{6,7} and 18 trials,^{11-28,57-59} reported on first-line systemic treatment compared with other treatments (the CheckMate 214 trial was reported in three papers,^{13,14,56} the KEYNOTE 426 trial was reported in three papers,^{16,57,59} and the IMmotion151 trial was reported in two papers^{17,58}); 21 papers, comprising one systematic review,⁷ and 20 trials,³⁰⁻⁴⁹ reported on second-line or greater systemic treatment compared with other treatments (the METEOR trial was reported in two papers^{33,34} as was the AXIS trial^{31,44}); three papers, comprising one systematic review⁸ and two trials,^{50,51} reported on metastases-directed treatments; two trials^{52,53} reported on bone metastases-directed therapy; one trial⁵⁴ reported on brain metastases-directed therapy; and one trial⁵⁵ reported on sarcomatoid-directed therapy. Three trials (CheckMate 214^{13,14} [first-line systemic treatment compared with other options and sarcomatoid-directed treatment options], CheckMate 025³⁰ [second-line systemic treatment compared with other treatment options and bone

metastases-directed treatment options], and METEOR^{33,34} [second-line systemic treatment compared with other treatment options and bone metastases-directed treatment options]) provided outcome data for more than one comparison of interest. See [Tables 1-8](#) for results. See [Data Supplement 1](#) for details on the included studies.

Quality of the included evidence was assessed via the AGREE2⁶⁰ instrument for the practice guideline, AMSTAR 2⁶¹ for the systematic reviews, and the certainty of evidence for outcomes reported in the clinical trials and the retrospective review was assessed using the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) instrument using MAGIcapp software.⁶² Overall, the included evidence was of high quality (except for outcomes from the retrospective review, which were rated as low according to GRADE because of nonrandomized design). Full details of the quality of evidence assessments can be found in the [Data Supplement 2: Study Quality Assessment](#).

RECOMMENDATIONS

Clinical Question 1

How is metastatic clear cell renal cell carcinoma defined and how is it diagnosed?

Recommendation 1.1. The diagnosis of metastatic ccRCC should ideally involve comparison of tissue acquired outside the site of primary disease to the primary histology. Histologic evaluation should include common markers of ccRCC including paired box gene 8 (PAX8) and carbonic anhydrase IX (CAIX; Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 1.2. Radiographic diagnosis of metastatic ccRCC may be applied in selected circumstances, such as settings in which a prior diagnosis of renal cell carcinoma has been established, when metastatic tissue is not readily accessible by biopsy, or when RECIST 1.1 measurable disease is evident, especially within a year of the initial diagnosis (Type: Consensus based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Weak).

Literature review and analysis. ccRCC is expected to be the dominant subtype in the metastatic setting.⁶³ The definition of diagnosis of metastatic ccRCC is ideally made by comparing the pathologic features of the metastatic and primary renal tumors, with the caveat that the metastatic tumor may have originated from an area of the primary tumor that was not sampled for histopathologic evaluation.⁶⁴ In scenarios when diagnostic material from the primary tumor may not be available, a definitive diagnosis of metastatic ccRCC requires strict morphologic assessment and the aid of ancillary studies such as immunohistochemistry or molecular assays.^{65,66} The presence of non-clear cell features in a metastatic sample (eosinophilic cytoplasm, or rhabdoid or sarcomatoid morphology) makes

TABLE 1. Main Findings From Systematic Reviews and/or Meta-Analyses

Study	Publication Type	Evidence Base	Main Findings	Quality Score (instrument)
Cytoreductive nephrectomy				
Bex et al ⁵	Guideline	Two RCTs	Immediate CN should no longer be the standard of care for patients with intermediate- and poor-risk metastatic renal clear cell carcinoma when medical treatment is required. Instead, the treatment offered should be systemic therapy using one or more of the recommended options (ie, sunitinib, nivolumab, and ipilimumab).	Rigor of development score: 83% (two raters)
First-line systemic treatment				
Haaland et al ⁶	Systematic review	Four RCTs	Bevacizumab plus interferon, sunitinib, and pazopanib are acceptable first-line options. Interferon alone should not be considered for first-line treatment.	31% (5/16)
Coppin et al ⁷	Systematic review	15 studies involving 5,587 patients testing the antivasculature epithelial growth factor agents bevacizumab, sorafenib, sunitinib, pazopanib, tivozanib, and axitinib	Agents targeting VEGF and mTOR pathways improve PFS in both first-line and second-line settings. These treatments should not be considered curative. No placebo-controlled trial has detected HRQoL benefits.	33% (4/12)
Second- or later-line systemic treatment				
Coppin et al ⁷	Systematic review	15 studies involving 5,587 patients testing the antivasculature epithelial growth factor agents bevacizumab, sorafenib, sunitinib, pazopanib, tivozanib, or axitinib	Agents targeting VEGF and mTOR pathways improve PFS in both first-line and second-line settings. These treatments should not be considered curative. No placebo-controlled trial has detected HRQoL benefits.	33% (4/12)
Metastases-directed therapy				
Zaorsky et al ⁸	Systematic review	28 studies including 1,602 mutually exclusive patients (679 extracranial/923 intracranial) and 3,892 lesions (1,159 extracranial/2,733 intracranial)	Stereotactic radiotherapy is safe and efficacious for RCC oligometastases, with 1-year local control at 90% and any significant toxicity at 1%.	56% (9/16)

Abbreviations: CN, cytoreductive nephrectomy; HRQoL, health-related quality of life; mTOR, mammalian target of rapamycin; PFS, progression-free survival; RCT, randomized controlled trial; VEGF, vascular endothelial growth factor.

TABLE 2. Trial Outcomes: Cytoreductive Nephrectomy Versus Other

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Méjean et al ⁹ CARMENA NCT00930033 2009-2017	Nephrectomy followed by sunitinib	OS Median follow-up of 50.9 months	226	13.9 months	7.2 months	NR	32.8	High
	Sunitinib alone		224	18.4 months HR = 0.89; 95% CI, 0.71 to 1.10; P > .05	8.3 months HR = 0.82; 95% CI, 0.67 to 1.00; P > .05	NR	42.7	
Bex et al ¹⁰ SURTIME NCT01099423 2010-2016	Sunitinib followed by cytoreductive nephrectomy	PFS Median follow-up of 39 months	49	32.4 months	NR	NR	58.3	High
	Cytoreductive nephrectomy followed by sunitinib		50	15.0 months HR = 0.57; 95% CI, 0.34 to 0.95; P = .03	NR HR = 0.88; 95% CI, 0.56 to 1.37; P = .57	NR	52.2	

Abbreviations: GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; NR, not reported; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; PRO, patient-reported outcome; QoL, quality of life.

establishing a clear cell renal origin difficult but may be solved in most cases by ancillary studies. PAX8 expression would support a renal origin,⁶⁷ and diffuse CAIX expression would support an origin from ccRCC.⁶⁸

Clinical interpretation. Tissue analysis of cells obtained outside the area of the primary tumor, which meets these criteria of correlation to the primary tumor cells or with PAX8 and or CAIX expression, sets the standard for diagnosing metastatic disease with a high level of evidence.

As a matter of practice, metastatic disease is sometimes declared based purely on radiographic findings. Widespread lesions on any imaging modality in the setting of a confirmed primary ccRCC, particularly of higher stage, may be interpreted as stage IV. Generally, lesions measurable using RECIST 1.1 criteria should be evaluated⁶⁹ and reported. Nontarget lesions, such as pleural or peritoneal fluid, should generally be sampled and confirmed to contain malignance cells before using this finding as a criterion to establish metastatic disease. Notably, the temporal association of metastatic findings is also an important factor to consider. The International Metastatic RCC Database Consortium (IMDC) model applies evidence of metastatic findings prompting intervention less than one year from initial diagnosis as a major risk criterion.⁷⁰ Thus, review of metastatic criterion should be considered carefully in determining plans for treatment of metastatic ccRCC.

Clinical Question 2

What is the role of cytoreductive nephrectomy in metastatic clear cell renal cell carcinoma?

Recommendation 2.1. Select patients (see *Practical Information*) with metastatic ccRCC may be offered CN (Type:

Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Practical Information: Select patients include those with optimally one IMDC risk factor who can have a significant majority of their tumor burden removed at the time of surgery.

Literature review and analysis. CN followed by systemic cytokine therapy with interferon alpha (IFN- α) was shown to provide a survival benefit over surgery alone by two prospective, randomized cooperative group trials in 2001.^{71,72} Combined and long-term analyses continued to favor CN + IFN- α versus surgery alone.^{73,74} Retrospective and National Cancer Database (NCDB) studies found OS benefit for CN plus systemic therapy in the vascular endothelial growth factor (VEGF) tyrosine kinase inhibitor (TKI) era.^{75,76} CN plus systemic targeted therapies also demonstrated an overall mortality benefit compared with surgery alone on meta-analysis.⁷⁷

However, the CARMENA trial, which was a prospective, randomized, phase III noninferiority study of CN followed by sunitinib versus sunitinib alone, found that the sunitinib alone arm had a longer median OS (18.4 months, 95% CI, 14.7 to 23.0 v 13.9 months, 95% CI, 11.8 to 18.3) and median PFS (8.3 months, 95% CI, 6.2 to 9.9 v 7.2 months, 95% CI, 6.7 to 8.5).⁹ Critiques of this study include that it was underpowered, having only enrolled 450 of planned 576 subjects, and slow to accrue over eight years across 79 centers. There were also many patients with IMDC poor-risk features and a high burden of disease.

Post hoc analysis of CARMENA looking at the OS of patients allocated to sunitinib only who underwent secondary

TABLE 3. Trial Outcomes: Systemic Treatment in the First Line

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival			Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS	QoL		
Motzer et al ¹¹ CLEAR NCT02811861 2016-2019	Lenvatinib plus pembrolizumab	PFS Median follow-up for OS of 26.6 months	355	79.2% Median, not reached	23.9 months	NR	82.4	High
	Lenvatinib plus everolimus		357	66.1% Median, not reached	14.7 months	NR	83.1	
	Sunitinib		357	70.4% Median, not reached	9.2 months	NR	71.8	
Choueiri et al ¹² CheckMate 9ER NCT03141177 2017-2019	Nivolumab plus cabozantinib	PFS Median follow-up for OS of 18.1 months	323	85.7% at 12 months Median, not reached	16.6 months	FKSI-19 58.7 ± 10.6	75.3	High
	Sunitinib		328	75.6% at 12 months Median, not reached HR = 0.60; 98.89% CI, 0.40 to 0.89; <i>P</i> = .001	8.3 months HR = 0.51; 95% CI, 0.41 to 0.64; <i>P</i> < .001	FKSI-19 58.4 ± 9.9	70.6	
Motzer et al ¹³ and Tannir et al ¹⁶ (efficacy outcomes), Cella et al ¹⁴ (QoL outcomes only) CheckMate 214 NCT02231749 2014-2016	Nivolumab plus ipilimumab followed by nivolumab	OS, PFS, ORR Minimum follow-up of 60 months for efficacy outcomes and 25.2 months for QoL outcomes	Intermediate- risk/poor-risk 425	Intermediate-risk/poor- risk 52% 47 months	Intermediate-risk/ poor-risk 33% 11.6 months	Intermediate-risk/poor- risk Mean change from baseline PROs at week 103: 4.00	Treatment- related 47.3	High
			Favorable-risk 125	Favorable-risk 70% Median not reached	Favorable-risk 27% 17.0 months	Favorable-risk NR		
	Sunitinib		Intermediate- risk/poor-risk 422	Intermediate-risk/poor- risk 39% 26.6 months 48 months: HR = 0.66; 95% CI, 0.55 to 0.80; <i>P</i> < .0001 60 months: HR = 0.46; 95% CI, 0.29 to 0.71; <i>P</i> = .0004	Intermediate-risk/ poor-risk 16% 8.3 months 48 months: HR = 0.75; 95% CI, 0.62 to 0.90; <i>P</i> < .0015 60 months: HR = 0.50; 95% CI, 0.32 to 0.80; <i>P</i> = .0036	Intermediate-risk/poor- risk Mean change from baseline PROs at week 103: -3.14 <i>P</i> < .0001 in favor of nivolumab plus ipilimumab followed by nivolumab	Treatment- related 64.1	
			Favorable-risk 124	Favorable-risk 73% Median not reached HR = 1.19; 95% CI, 0.77 to 1.85; <i>P</i> > .04383	Favorable-risk 33% 28.8 months HR = 1.65; 95% CI, 1.16 to 2.35; <i>P</i> = .0049	Favorable-risk NR		

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TABLE 3. Trial Outcomes: Systemic Treatment in the First Line (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Choueiri et al ¹⁵ JAVELIN Renal 101 NCT02684006 2016-2017	Avelumab plus axitinib	OS, PFS Minimum follow-up of 13 months	442	Not reached	13.3 months	NR	NR	High
	Sunitinib		444	Not reached HR = 0.80; 95% CI, 0.616 to 1.027; <i>P</i> = .0392	8.0 months HR = 0.69; 95% CI, 0.574 to 0.825; <i>P</i> < .0001	NR	NR	
Rini et al, ¹⁶ and Powles et al, ⁵⁹ and Rini et al ⁵⁷ KEYNOTE-426 NCT02853331 2016-2018	Pembrolizumab plus Axitinib	OS, PFS Median follow-up of 42.8 months	432	45.7 months	15.7 months	NR	75.8	High
	Sunitinib		429	40.1 months HR = 0.73; 95% CI, 0.60 to 0.88; <i>P</i> < .001 in favor of pembrolizumab plus axitinib	11.1 months HR = 0.68; 95% CI, 0.58 to 0.80; <i>P</i> < .0001	NR	70.6	
Rini et al ¹⁷ and Motzer et al ⁵⁸ IMmotion151 NCT02420821 2015-2016	Atezolizumab plus bevacizumab	PFS, OS Median follow-up of 15 months	454	57% 33.6 months	11.2 months	NR	Treatment- related 40.4	High
	Sunitinib		461	58% 34.9 months HR = 0.93; 95% CI, 0.76 to 1.14; <i>P</i> = .4751	8.4 months HR = 0.83; 95% CI, 0.70 to 0.97; <i>P</i> < .0219	NR	Treatment- related 53.8	
Cirkel et al ¹⁸ ROPETAR NCT01408004 2012-2014	Pazopanib and everolimus	PFS Minimum follow-up of 12 months	52	35 months	20.2 months	NS FKSI-DRS symptom scale, physical functioning, and overall QoL according to QLQ- C30 between treatment arms	43.1	High
	Pazopanib		49	18.5 months HR (stratified for MSKCC risk categories) = 0.90; 95% CI, 0.51 to 1.58; <i>P</i> > .05 (NR)	14.5 months HR = 1.02; 95% CI, 0.59 to 1.76; log- rank <i>P</i> = .86	NS FKSI-DRS symptom scale, physical functioning, and overall QoL according to QLQ-C30 between treatment arms	49.0 <i>P</i> = .69	
Rini et al ¹⁹ IMPRINT NCT01265901 2010-2012	Sunitinib plus IMA901	OS Median follow-up of 33.27 months	204	33 months	15.22 months	NR	57.4	High
	Sunitinib		135	Not reached HR = 1.34; 95% CI, 0.96 to 1.86; <i>P</i> = .087	15.12 months Log-rank <i>P</i> = .74 HR = 1.05; 95% CI, 0.77 to 1.43	NR	46.9	

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TABLE 3. Trial Outcomes: Systemic Treatment in the First Line (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Choueiri et al ²⁰ Alliance A031203 CABOSUN NCT01835158 2013-2015	Cabozantinib	PFS	79	30.3 months	8.2 months	NR	66.7	High
	Sunitinib	Median follow-up of 21.4 months	78	21.8 months HR = 0.80; 95% CI, 0.50 to 1.26; <i>P</i> > .05 (NR)	5.6 months HR (for progression or death) = 0.66; 95% CI, 0.46 to 0.95; (one-sided) <i>P</i> = .012	NR	68.1	
Rini et al ²¹ INTORACT NCT00631371 2008-2010	Temsirolimus/ bevacizumab	PFS	400	25.8 months	9.1 months	FKSI-15: 43.3 FKSI-DRS: 29.2	80	High
	IFN/bevacizumab	Median follow-up NR	391	25.5 months HR = 1.0; 95% CI, 0.9 to 1.3; <i>P</i> = .6	9.3 months HR = 1.1; 95% CI, 0.9 to 1.3; <i>P</i> = .8	FKSI: 41.5; <i>P</i> = .002 FKSI-DRS: 28.0; <i>P</i> < .001 EQ-5D, EQ-VAS NS	76	
Motzer et al ²² COMPARZ NCT00720941 2008-2011	Pazopanib	PFS	557	28.4 months	8.4 months	Superior	Grade 3: 59.0 Grade 4: 15.3	High
	Sunitinib	Median follow-up NR	553	29.3 months HR = 0.91; 95% CI, 0.76 to 1.08; <i>P</i> = .28	9.5 months HR = 1.05; 95% CI, 0.90 to 1.22; <i>P</i> > .05 (NR)	Inferior	Grade 3: 56.8 Grade 4: 16.6	
Hutson et al ²³ A4061051 NCT00920816 2010-2011	Axitinib	PFS	192	NR	10.1 months	NR	All-cause serious adverse effects 33.9	High
	Sorafenib	Median follow-up NR	96	NR	6.5 months HR = 0.77; 95% CI, 0.56 to 1.05; <i>P</i> = .038	NR	All-cause serious adverse effects 25	
Rini et al ²⁴ NCT00467025 2007-2008	AMG 386 (10 mg/kg qw)/sorafenib	PFS	50	Not reached	9 months	NR	66	High
	AMG 386 (3 mg/kg qw)/sorafenib	Median follow-up of 17 months	51	29.2 months	8.5 months	NR	73	
	Placebo/sorafenib		51	27.1 months <i>P</i> = NR	9.0 months HR (arms A and B compared with C) = 0.88; 95% CI, 0.60 to 1.30; <i>P</i> = .52	NR	86	

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TABLE 3. Trial Outcomes: Systemic Treatment in the First Line (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Rini et al ²⁵ NCT00072046 CALGB 90206 2003-2005	Bevacizumab plus IFN- α	OS Median follow-up of 46.2 months	369	18.3 months	NR	NR	80.1	High
	IFN- α monotherapy		363	17.4 months HR = 0.86; 95% CI, 0.73 to 1.01; stratified log- rank, $P = .069$	NR	NR	62.5	
Jonasch et al ²⁶ 2005-2007	Sorafenib	ORR, safety Median follow-up of 19.7 months	40	Not reached	7.4 months	NR	NR	High
	Sorafenib and low- dose IFN- α		40	27 months HR = 1.945; 95% CI, 0.837 to 4.517; $P = .1219$	7.6 months HR, 0.848; 95% CI, 0.507 to 1.418; $P = .5289$	NR	NR	
Motzer et al ²⁷ NCT00098657 , NCT00083889 2004-2005	Sunitinib	PFS Median follow-up NR	375	26.4 months	11 months	NR	NR	High
	IFN- α		375	21.8 months HR = 0.821; 95% CI, 0.673 to 1.001; $P = .051$	5 months HR = 0.539; 95% CI, 0.451 to 0.643; $P < .001$	NR	NR	
Hudes et al ²⁸ ARCC NCT00065468 2003-2005	IFN- α alone	OS Median follow-up NR	207	7.3 months	1.9 months	NR	78	High
	Temsirolimus alone		209	10.9 months IFN- α v temsirolimus: HR = 0.73; 95% CI, 0.58 to 0.92; $P = .008$	3.8 months	NR	67	
	Interferon plus temsirolimus		210	8.4 months Temsirolimus v IFN- α plus temsirolimus: HR = 0.96; 95% CI, 0.76 to 1.20; $P = .70$	3.7 months	NR	87	

Abbreviations: EQ-5D, The European Quality of Life Group Patient-Reported Outcomes measure; EQ-VAS, The European Quality of Life Group Visual Analog Scale; FKS-15, Functional Assessment of Cancer Therapy Kidney Cancer Symptom Index; FKS-DRS, Functional Assessment of Cancer Therapy Kidney Cancer Symptom Index—Disease-Related Symptoms; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; IFN- α , interferon alpha; MSKCC, Memorial Sloan Kettering Cancer Center; NR, not reported; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; QLQ-C30, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire; QoL, quality of life.

TABLE 4. Trial Outcomes: Systemic Treatment in the Second Line or Beyond

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Iacovelli et al ²⁹ Retrospective review	Early cabozantinib	PFS, OS, ORR	39	Not reached	Not reached	NR	NR	Low
	Late cabozantinib	Median follow-up of 9.4 months	45	14.7 months HR = 0.31; 95% CI, 0.11 to 0.92; <i>P</i> = .035	11.1 months HR = 0.50; 95% CI, 0.22 to 1.11; <i>P</i> = .089	NR	NR	
Motzer et al ³⁰ CheckMate 025 NCT01668784 2012-2014	Nivolumab	OS Median follow-up of 72 months	410	25.8 months	4.2 months	Superior	Treatment-related 21.4	High
	Everolimus		411	19.7 months HR = 0.73; 95% CI, 0.62 to 0.85; <i>P</i> < .0001	4.5 months HR = 0.84; 95% CI, 0.72 to 0.99; <i>P</i> = .0331	Inferior	Treatment-related 36.8	
Bracarda et al ³¹ Post hoc analysis of the AXIS trial NCT00678392	First-line sunitinib, then second-line axitinib	PFS Median follow-up of 26.5 months for axitinib; 25.7 months for sorafenib	194	32.4 months for MSKCC favorable-/intermediate-risk, nonbulky disease, and no bone or liver metastases	13.9 months for MSKCC favorable-/intermediate-risk, nonbulky disease, and no bone or liver metastases	NR	72.9	High
	First-line sunitinib, then second-line sorafenib		195	35.0 months for MSKCC favorable-/intermediate-risk, nonbulky disease, and no bone or liver metastases HR = 0.902; 95% CI, 0.457 to 1.780; two-sided <i>P</i> = .7661	4.7 months for MSKCC favorable-/intermediate-risk, nonbulky disease, and no bone or liver metastases HR = 0.476; 95% CI, 0.263 to 0.863; two-sided <i>P</i> = .0126	NR	67.9	
Voss et al ³² DART NCT01727336 2014-2017	Axitinib plus dalantercept	PFS Median follow-up of 17.8 months	59	Median not reached	6.8 months	NR	59	High
	Axitinib plus placebo		65	Median not reached HR = 1.39; 95% CI, 0.70 to 2.77; <i>P</i> = .349	5.6 months HR = 1.11; 95% CI, 0.71 to 1.73; <i>P</i> = .670	NR	64	
Motzer et al ³³ Choueiri et al ³⁴ (PFS data only) METEOR NCT01865747 2013-2014	Cabozantinib	PFS	330	21.4 months	7.4 months	NR	71.3	High
	Everolimus	Median follow-up of 18.7 months (PFS, cabozantinib), 18.8 months (PFS, everolimus); 28 months (OS) All patients had received at least one prior VEGFR tyrosine kinase inhibitor	328	17.1 months HR = 0.70; 95% CI, 0.58 to 0.85; <i>P</i> = .0002	3.9 months HR = 0.51; 95% CI, 0.41 to 0.62; <i>P</i> < .0001	NR	60.9	

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TABLE 4. Trial Outcomes: Systemic Treatment in the Second Line or Beyond (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Voss et al ³⁵ CRLX101-208 NCT02187302 2014-2016	CRLX101 with bevacizumab	PFS All patients had received two or three prior lines of treatment	50 analyzed of 56 randomly assigned	16.1 months	3.7 months	NR	Treatment- related adverse effects 50.9	High
	Standard of care (eg, approved TKIs such as axitinib, sorafenib, and sunitinib, or mTOR inhibitors such as everolimus or temsirolimus as long as it was not given in previous lines of therapy)		52 analyzed of 59 randomly assigned	16.4 months HR = 1.21; 95% CI, 0.63 to 2.36; <i>P</i> = .718	3.9 months HR = 1.25; 95% CI, 0.78 to 2.01; <i>P</i> = .822	NR	Treatment- related adverse effects 46.4	
Powles et al ³⁶ COSAK NCT00942877 2010-2012	Cediranib and saracatinib	PFS Median follow-up of 12 months	69	14.2 months	3.9 months	NR	NR	High
	Cediranib and placebo	Eligible patients demonstrated disease progression after ≥ 1 lines of VEGF- targeted therapy	69	10.0 months HR = 1.28; 1.00 to 1.63; <i>P</i> > .05	5.4 months HR = 1.18; 0.94 to 1.48; <i>P</i> > .05	NR	NR	
Powles et al ³⁷ 2013-2014	AZD2014	PFS	26	4.9 months	1.8 months	NR	35	High
	Everolimus	Eligible patients demonstrated disease progression after ≥ 1 lines of VEGF- targeted therapy	23	Not reached HR = 4.3; 95% CI, 1.1 to 16.5; <i>P</i> = .02	4.6 months HR = 2.8; 95% CI, 1.2 to 6.5; <i>P</i> = .01	NR	48	
Powles et al ³⁸ 2011-2012	Apatolisib	PFS Median follow-up of 16.6 months	42	16.5 months	3.7 months	NR	73.8	High
	Everolimus	Eligible patients demonstrated disease progression after ≥ 1 lines of VEGF- targeted therapy	43	22.8 months HR = 1.77; 95% CI, 0.97 to 3.24; <i>P</i> = .06	6.1 months HR = 2.12; 95% CI, 1.23 to 3.63; <i>P</i> < .01	NR	44.2	
Flaherty et al ³⁹ ECOG-ACRIN E2804 NCT00378703 2007-2010	Bevacizumab	PFS	89	28.6 months	7.5 months	NR	44	High
	Bevacizumab/ temsirolimus	Median follow-up NR	91	24.7 months	7.6 months	NR	77	
	Bevacizumab/ sorafenib		90	27.5 months	9.2 months	NR	82	
	Sorafenib/temsirolimus		91	24.3 months <i>P</i> = NS for any comparison	7.4 months <i>P</i> = NS for any comparison	NR	84	

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TABLE 4. Trial Outcomes: Systemic Treatment in the Second Line or Beyond (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival			Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS	QoL		
Motzer et al ⁴⁰ E7080-G000-205 NCT01136733 2012-2013	Lenvatinib	PFS	52	18.4 months	7.4 months	NR	78.8	High
	Everolimus	Median follow-up of 17.8 months (lenvatinib), 16.5 months (everolimus), and 18.5 months (lenvatinib plus everolimus) for survival	50	17.5 months	5.5 months	NR	50.0	
	Lenvatinib plus everolimus	Eligible patients demonstrated disease progression after VEGF-targeted therapy	51	25.5 months Lenvatinib plus everolimus compared with single-agent everolimus; HR = 0.55; 95% CI, 0.30 to 1.01; <i>P</i> = .062 Single-agent lenvatinib compared with single-agent everolimus, HR = 0.74,95% CI, 0.42 to 1.31; <i>P</i> = .29 Lenvatinib plus everolimus compared with single-agent lenvatinib, HR = 0.74,95% CI, 0.40 to 1.36; <i>P</i> = .30	14.6 months Lenvatinib plus everolimus compared with single-agent everolimus; HR = 0.40; 95% CI, 0.24 to 0.68; <i>P</i> = .0005 Single-agent lenvatinib compared with single-agent everolimus, HR = 0.61, 95% CI, 0.38 to 0.98; <i>P</i> = .048 Lenvatinib plus everolimus compared with single-agent lenvatinib, HR = 0.66, 95% CI, 0.39 to 1.10; <i>P</i> = .12	NR	70.6	
Motzer et al ⁴¹ CTKI258A2302 NCT01223027 2011-2012	Dovitinib	PFS Median follow-up NR All patients had received a prior line of VEGF targeted therapy and an mTOR inhibitor line	284	11.1 months	3.7 months	NR	Treatment-related adverse effects 47.5	High
	Sorafenib		286	11.0 months HR = 0.96; 95% CI, 0.75 to 1.22; <i>P</i> = NS (NR)	3.6 months HR = 0.86; 95% CI, 0.72 to 1.04; one-sided <i>P</i> = .063	NR	Treatment-related adverse effects 39.4	
Hutson et al ⁴² INTORSECT NCT00474786 2007-2011	Temsirolimus	PFS	259	12.3 months	4.3 months	NR	70	High
	Sorafenib	Median follow-up of 9.2 months All patients must have received first-line sunitinib	253	16.6 months HR = 1.31; 95% CI, 1.05 to 1.63; two-sided <i>P</i> = .01	3.9 months HR = 0.87; 95% CI, 0.71 to 1.07; two-sided <i>P</i> = .19	NR	69	
Motzer et al ⁴³ TIVO-1 NCT01030783 2010	Tivozanib	PFS Median follow-up NR	260	28.8 months	11.9 months	FACT-G −2.83	61.4	High
	Sorafenib		257	29.3 months HR = 1.245; 95% CI, 0.954 to 1.624; <i>P</i> = .105	9.1 months HR = 0.797; 95% CI, 0.64 to 0.99; <i>P</i> = .042	−3.10 <i>P</i> = .805	69.6	
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TABLE 4. Trial Outcomes: Systemic Treatment in the Second Line or Beyond (continued)

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival			Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS	QoL		
Motzer et al ⁴⁴ AXIS NCT00678392 2008-2010	Axitinib	PFS	361	20.1 months	8.3 months	NR	NR	High
	Sorafenib	Median follow-up NR All patients had progressive disease after one approved systemic treatment	362	19.2 months HR = 0.969, 95% CI, 0.800 to 1.174; one- sided $P = .3744$	5.7 months HR = 0.656, 95% CI, 0.552 to 0.779; one- sided $P < .0001$	NR	NR	
Liu et al ⁴⁵ 2005-2008	iv-CIKs	OS	74	46 months	12 months	NR	NR	High
	sc-IL-2 + sc-IFN- α -2a	Median follow-up of 27 months	74	19 months HR = NR; $P < .001$	8 months HR = NR; $P = .024$	NR	NR	
Mulders et al ⁴⁶ D8480C00030 NCT00423332 2007	Cediranib	Change from baseline in tumor size after 12 weeks	53	NR	12.1 months	NR	75	High
	Placebo	Approximate follow-up of 12 months	18	NR	2.8 months HR = 0.45; 90% CI, 0.26 to 0.78; $P = .017$	NR	NR	
Nosov et al ⁴⁷ 2007-2008	Tivozanib (unselected patients)	ORR, PFS was a secondary outcome	78	NR	NR	NR	NR	High
	Randomly assigned to tivozanib	Median follow-up NR	61	NR	10.3 months	NR	NR	
	Randomly assigned to placebo		57	NR	3.3 months $P = .010$	NR	NR	
	No treatment (unselected patients)		76	NR	NR	NR	NR	
Motzer et al ⁴⁸ CRAD001C2240 NCT00410124 2006-2007	Everolimus	PFS	272	Not reached	4.0 months	NR	NR	High
	Placebo	Median follow-up NR All patients had received a prior line of VEGF targeted therapy	138	8.8 months HR = 0.83, 95% CI, 0.50 to 1.37; $P = .23$	1.9 months HR = 0.30, 95% CI, 0.22 to 0.40; $P < .0001$	NR	NR	
Escudier et al ⁴⁹ 11,213 NCT00073307 2003-2005	Sorafenib	OS	451	Not reached	5.5 months	NR	NR	High
	Placebo	Median follow-up 6.6 months All patients had progressed after one systemic treatment within the previous 8 months	452	14.7 months HR = 0.72; 95% CI, 0.54 to 0.94; $P = .02$	2.8 months HR = 0.44; 95% CI, 0.35 to 0.55; $P < .001$	NR	NR	

Abbreviations: FACT-G, Functional Assessment of Cancer Therapy-General; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; IFN- α , interferon alpha; MSKCC, Memorial Sloan Kettering Cancer Center; mTOR, mammalian target of rapamycin; NR, not reported; NS, not significant; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; QoL, quality of life; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor; VEGFR, vascular endothelial growth factor receptor.

TABLE 5. Trial Outcomes: Metastases-Directed Therapy

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		QoL	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Choueiri et al ⁵⁰ KEYNOTE-564 NCT03142334	Adjuvant pembrolizumab	DFS Median time from random assignment to the data cutoff date was 24.1 months	496	96.6%	77.3%	NR	32.4	
	Placebo		498	93.5% HR = 0.54; 95% CI, 0.30 to 0.96	68.1% HR = 0.68; 95% CI, 0.53 to 0.87; P = .002	NR	17.1	
Procopio et al ⁵¹ RESORT NCT01444807 2012-2017	Radical metastasectomy followed by sorafenib	RFS Median follow-up of 38 months	33 (32)	Not reached	RFS: 21 months	NR	22	High
	Radical metastasectomy followed by observation		36	Not reached	RFS: 37 months Log-rank test, P = .404	NR	3	

Abbreviations: DFS, disease-free survival; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; NR, not reported; OS, overall survival; PFS, progression-free survival; QoL, quality of life; RFS, recurrence-free survival.

nephrectomy was interesting.⁷⁸ A total of 40 (18%) patients in the sunitinib alone group (n = 224) underwent secondary CN, and of these 40 patients, 31% resumed sunitinib treatment following surgery. Patients with secondary CN had a significantly longer OS (median 48.5 months) than those who had no subsequent CN

TABLE 6. Trial Outcomes: Bone Metastases-Directed Therapy

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		Skeletal- Related Events (%)	Grade 3-5 Adverse Events	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Escudier et al ⁵² METEOR NCT01865747 2013-2014	Cabozantinib	PFS Median follow-up NR	77 with bone metastases	20.1 months	7.4 months	23	73% (with bone metastases) 71% (without bone metastases)	High
	Everolimus		65 with bone metastases	12.1 months HR = 0.54; 95% CI, 0.34 to 0.84; P < .05 (NR)	2.7 months HR = 0.33; 95% CI, 0.21 to 0.51; P < .05 (NR)	29	51% (with bone metastases) 62% (without bone metastases)	
Escudier et al ⁵³ CheckMate 025 NCT01668784 2012-2014	Nivolumab	OS Median follow-up of 22 months	76 with bone metastases of 410 total	18.5 months	NR	NR	Treatment- related adverse effects 17% (with bone metastases) 19% (all patients)	High
	Everolimus		70 with bone metastases of 411 total	13.8 months HR = 0.72; 95% CI, 0.47 to 1.09; P = NS	NR	NR	Treatment- related adverse effects 19% (with bone metastases) 37% (all patients)	

Abbreviations: GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; NR, not reported; OS, overall survival; PFS, progression-free survival; QoL, quality of life.

TABLE 7. Trial Outcomes: Brain Metastases–Directed Therapy

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		Intracranial Response (%)	Grade 3-5 Adverse Events (%)	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Flippot et al ⁵⁴ GETUG-AFU 26 NIVOREN NCT03013335 2016-2017	Nivolumab	Intracranial response rate in cohort A Median follow-up was 23.6 months in cohort A and 20.2 months in cohort B	39 patients with previously untreated brain metastases	67% at 12 months	Intracranial PFS: 2.7 months	12	Treatment- related adverse effects 10	High
			34 patients whose brain metastases underwent prior therapy	59% at 12 months	Intracranial PFS: 4.8 months HR = 2.04; 95% CI, 1.08 to 3.83; P > .05 (NR)	NR	Treatment- related adverse effects 15	

Abbreviations: GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; NR, not reported; OS, overall survival; PFS, progression-free survival; QoL, quality of life.

(median 15.7 months, hazard ratio [HR] = 0.34; 95% CI, 0.22 to 0.54).

Similarly, SURTIME compared immediate versus deferred CN in patients with synchronous metastatic renal cell carcinoma (mRCC) receiving sunitinib.¹⁰ Participants were randomly assigned 1:1 to immediate CN followed by sunitinib versus sunitinib ×3 cycles followed by CN followed by sunitinib. PFS was the initial primary end point with a target accrual of 458 patients. However, because of poor accrual after 3 years, the independent data monitoring committee endorsed making the intention-to-treat (ITT) 28-week progression-free rate the primary end point and the accrual target was decreased to 98 patients. OS, adverse events (AEs), and postoperative progression remained secondary end points. PFS was not significantly different between the two groups (HR = 0.88; 95% CI, 0.56 to 1.37; *P* = .57), whereas deferred CN showed an improved OS (HR = 0.57; 95% CI, 0.34 to 0.95; *P* = .03).

SURTIME also demonstrated the feasibility of CN following systemic TKI therapy.⁷⁹ AEs related to surgery (all grades) in the immediate and deferred arms (per protocol) occurred in 52% and 53% of patients who underwent CN, respectively (difference in rates 0.77; 95% CI, 20.4 to 21.8). The

authors concluded that CN can be a morbid operation, but that neoadjuvant TKI did not increase operative risk.

Since CARMENA and SURTIME were completed, checkpoint inhibitor–based therapy has become a first-line treatment option for mRCC. A NCDB analysis of 391 surgical candidates diagnosed with clear cell mRCC treated with immunotherapy with or without CN and no other systemic therapies was recently published.⁸⁰ The primary outcome was OS stratified by the receipt of CN (CN plus immunotherapy v immunotherapy alone). Secondary outcomes included OS stratified by the timing of CN, pathologic findings, and perioperative outcomes. Of 391 patients, 221 (56.5%) received CN plus immunotherapy and 170 (43.5%) received immunotherapy only. CN plus immunotherapy had a longer OS than immunotherapy alone (HR = 0.23; 95% CI, 0.15 to 0.3; log-rank: *P* < .001).

Clinical interpretation. CN remains a treatment option for selected patients with synchronous mRCC.^{81,82} It can also be a palliative option for hematuria or pain.⁸³ CN should be considered for patients with one IMDC risk factor (intermediate risk disease).⁸⁴ Although a definitive prognostication system for CN is lacking, optimal candidates for CN have the significant majority of their tumor burden within

TABLE 8. Trial Outcomes: Sarcomatoid-Directed Therapy

Source	Intervention/ Comparisons	Primary End Points	No. of Patients Randomly Assigned (evaluated)	Survival		Response, %	Grade 3-5 Adverse Events, %	GRADE Certainty of Evidence
				Median or % OS	Median or % PFS			
Tannir et al ⁵⁵ CheckMate 214 NCT02231749 2014-2016	Nivolumab plus ipilimumab	OS, PFS, ORR (exploratory) Minimum follow- up of 42 months	74 patients with sarcomatoid features of 550 total	Not reached	26.5 months	ORR: 60.8 CR: 18.9	Treatment- related adverse effects 49.3	High
	Sunitinib		65 patients with sarcomatoid features of 546 total	14.2 months HR = 0.45; 95% CI, 0.3 to 0.7; P = .0004	5.1 months HR = 0.54; 95% CI, 0.33 to 0.86; P = .0093	ORR: 23.1 CR: 3.1	Treatment- related adverse effects 44.6	

Abbreviations: CR, complete response; GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; HR, hazard ratio; NR, not reported; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; QoL, quality of life.

the kidney, have good ECOG performance status, and do not have brain, bone, or liver metastases.⁸⁵ IMDC poor-risk patients are likely to benefit more from immediate systemic therapy, although delayed CN can be considered with good response to systemic treatment. As with many complex operations, CN is best performed by high-volume surgeons at high-volume centers as part of multidisciplinary consultation.^{86,87} Enrollment in a CN clinical trial should be encouraged.⁸⁸

Clinical Question 3

What are the preferred options for first-line systemic treatment of metastatic clear cell renal cell carcinoma?

Recommendation 3.1. Select patients with metastatic ccRCC (see *Practical Information*) may be offered an initial active surveillance strategy (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Practical Information: Select patients include those with IMDC favorable and intermediate risk, patients with limited or no symptoms related to disease, a favorable histologic profile, a long interval between nephrectomy and the development of metastasis, or with limited burden of metastatic disease.

Literature review and analysis. In a prospective phase II trial of active surveillance in 48 patients with treatment-naïve, asymptomatic metastatic RCC with a primary end point of time to initiation of systemic therapy, the median time on surveillance until initiation of systemic therapy was 14.9 months (95% CI, 10.6 to 25.0) with more IMDC risk factors and a higher number of metastatic sites associated with a shorter time on surveillance.^{89,90} Seven of the 48 patients received metastasis-directed therapy with surgery or radiation during the surveillance period. A second retrospective analysis using the Canadian Kidney Cancer information system (CKCis) defined active surveillance (cohort A) as patients (1) starting systemic therapy $\geq$ 6 months after diagnosis of mRCC (cohort A1), or (2) never receiving systemic therapy for mRCC with an OS of $\geq$ 1 year (cohort A2); and patients receiving systemic therapy $<$ 6 months after diagnosis of mRCC defined as receiving immediate systemic treatment (cohort B).⁹⁰ The 5-year OS probability was significantly greater for cohort A than for cohort B (70% v 33.6%; $P < .0001$). After adjusting for IMDC risk criteria and age, both OS (HR = 0.58; 95% CI, 0.47 to 0.70; $P < .0001$) and TTF (HR = 0.72; 95% CI, 0.60 to 0.85; $P = .0002$) were greater in cohort A1 compared with B. For cohort A1, the median time on active surveillance was 14.2 months (range, 6-71 months).

Clinical interpretation. A subset of carefully selected patients with mRCC may have a more indolent disease course such that an initial active surveillance strategy may be considered. Patients with lower IMDC risk score, limited number of metastatic sites, long intervals from nephrectomy

to developing metastatic disease, and low disease volume might be better candidates for this approach. Metastasis-directed therapy may be considered for select patients on surveillance. Close monitoring for disease progression with regular serial imaging, including bone and CNS imaging, is essential to this approach. The benefit of preserving QoL and delaying or avoiding treatment-related side effects versus the potential for disease progression should be discussed with the patient to incorporate their preferences in the decision-making process.

Recommendation 3.2. All patients with metastatic ccRCC who require systemic therapy in the first-line setting should undergo risk stratification into IMDC favorable (0), intermediate (1-2), and poor (3+) risk groups.³ Patients with intermediate- or poor-risk disease should be offered combination treatment with two immune checkpoint inhibitors (ICIs; ie, ipilimumab and nivolumab) or an ICI in combination with a vascular endothelial growth factor receptor (VEGFR TKI; Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Literature review and analysis. Major advances in the first-line management of metastatic ccRCC initially occurred with the development of VEGF-targeted therapy. In 2003, a randomized trial of the anti-VEGF antibody, bevacizumab, versus placebo in patients with progressive metastatic ccRCC and previous therapy with IL-2 (or contraindications to IL-2) demonstrated a significant prolongation of time to progression in patients receiving bevacizumab compared with placebo and heralded the paradigm shift to the use of VEGF targeted treatments for patients with advanced RCC.⁹¹ Subsequent studies using the VEGFR TKI sunitinib demonstrated improvement in response and survival outcomes compared with IFN- α with similar outcomes seen with the VEGFR TKI pazopanib leading to a new standard of care for patients with advanced ccRCC.^{22,92} In addition to VEGF TKIs, temsirolimus, an inhibitor of mammalian target of rapamycin kinase, demonstrated an improvement in OS compared with IFN- α in patients with poor-prognosis metastatic RCC.²⁸ In the A031203 trial, cabozantinib, an oral inhibitor of MET, AXL, and VEGFR2 demonstrated an improvement in median PFS compared with sunitinib (8.6 months, 95% CI, 6.8 to 14.0 v 5.3 months, 95% CI, 3.0 to 8.2) for cabozantinib versus sunitinib (HR = 0.48; 95% CI, 0.31 to 0.74; two-sided $P = .0008$) in patients with previously untreated advanced RCC (intermediate or poor risk by IMDC criteria).⁹³ More recently, phase III randomized trials using sunitinib as the de facto comparator arm have led to a new standard of care incorporating ICIs in the treatment of patients with advanced RCC.

CheckMate 214: Nivo plus Ipi versus sunitinib. CheckMate 214 was the pivotal phase III trial investigating the role of dual ICI therapy with nivolumab, an anti-programmed

death-1 (PD-1) agent in combination with ipilimumab, an anti-cytotoxic T-cell lymphocyte-4 (CTLA-4) agent in patients with previously untreated advanced renal cell carcinoma. In this study, a total of 1,096 patients were randomly assigned 1:1 to receive nivolumab plus ipilimumab (Nivo plus Ipi) or sunitinib. Nivolumab 3 mg/kg and ipilimumab 1 mg/kg were administered intravenously once every 3 weeks for four cycles followed by nivolumab 3 mg/kg once every 2 weeks. Sunitinib was administered 50 mg orally once daily, 4 weeks on and 2 weeks off in each 6-week cycle. Primary end points were OS, PFS, and objective response rate (ORR) in the IMDC intermediate- and poor-risk populations, and secondary end points were OS, PFS, and ORR in the ITT population, and safety. At a minimum study follow-up of 42 months in the primary efficacy population of intermediate-/poor-risk patients, Nivo plus Ipi improved OS probability (52% v 39%; HR = 0.66; 95% CI, 0.55 to 0.80; $P < .0001$) and PFS probability (33% v 16%; HR = 0.75; 95% CI, 0.62 to 0.90; $P < .0015$) when compared with sunitinib.¹³ In addition, higher ORR (42% v 26%; $P < .0001$) and complete response (CR) rates (10.1% v 1.4%) were seen with Nivo + Ipi when compared with sunitinib. Among favorable-risk patients, median OS was not reached in either cohort and the OS probabilities were comparable between the two arms (70% with Nivo plus Ipi v 73% with sunitinib [HR = 1.19; 95% CI, 0.77 to 1.85; $P > .05$; not reported; NR]). In the favorable-risk patients, ORR was higher with sunitinib (54% v 29%; $P < .0001$) but a higher proportion of patients achieved CR with Nivo plus Ipi (12.8% v 5.6%; $P = \text{NR}$). Forty-two-month PFS was higher in the Nivo + Ipi arm (32% v 20%) when compared with the sunitinib arm. Grade 3 or higher AEs were seen in 47.3% and 64.1% of the patients treated with Nivo plus Ipi and sunitinib, respectively.⁹⁴ Nivolumab plus ipilimumab was associated with fewer patient-reported symptoms and better health-related quality of life (HRQoL) than sunitinib.¹⁴

A 2022 abstract reporting on 5-year survival continued to favor Nivo plus Ipi for both OS (HR = 0.46; 95% CI, 0.29 to 0.71; $P = .0004$) and PFS (HR = 0.50; 95% CI, 0.32 to 0.80; $P = .0036$). ORR was higher with Nivo-ipi (61% v 23%; $P < .0001$), median duration of response was longer with Nivo-ipi (NR v 25 months), and more patients had CRs with Nivo-ipi (23% v 6%).⁵⁶ Nivo-ipi was superior to sunitinib independent of programmed death ligand-1 (PD-L1) expression, and the OS, PFS, and ORR benefits with Nivo plus Ipi were greater for patients with programmed death-ligand 1 (PD-L1) $\geq 1\%$. No change in safety data were reported in either arm.

JAVELIN Renal 101: Avelumab plus axitinib versus sunitinib.

The phase III JAVELIN Renal 101 trial enrolled 886 patients with treatment-naïve metastatic ccRCC and randomly assigned them 1:1 to avelumab 10 mg intravenously once every 2 weeks/kg plus axitinib 5 mg orally twice daily or sunitinib 50 mg once daily for 4 weeks on 2 weeks off every 6 weeks.^{15,95,96} The two independent primary end points were PFS and OS among patients with PD-L1–positive (PD-L1+) tumors.

Key secondary end points were OS and PFS in the overall population. Among the 560 patients with PD-L1–positive tumors (63.2%), the median PFS was 13.8 months with avelumab plus axitinib, compared with 7.2 months with sunitinib (HR for disease progression or death = 0.61; 95% CI, 0.47 to 0.79; $P < .001$); in the overall population, the median PFS was 13.8 months, compared with 8.4 months (HR = 0.69; 95% CI, 0.56 to 0.84; $P < .001$). Among the patients with PD-L1–positive tumors, the ORR was 55.2% with avelumab plus axitinib and 25.5% with sunitinib; at a median follow-up for OS of 11.6 months and 10.7 months in the two groups, 37 patients and 44 patients had died, respectively. Among patients who received axitinib in the combination group, 42.2% had at least one reduction in the dose of axitinib and 10.8% had at least one escalation in the dose of axitinib. Of the patients who received sunitinib, 42.6% had at least one dose reduction. AEs of grade 3 or higher during treatment occurred in 71.2% in the avelumab plus axitinib group and 71.5% in the sunitinib group. AEs that occurred during treatment led to discontinuation of both avelumab and axitinib in 7.6% who received the combination and led to discontinuation of sunitinib in 13.4% who received sunitinib. Stratified analyses demonstrated relatively consistent benefit of avelumab and axitinib, compared with sunitinib, across strata of IMDC risk groups. The OS data are pending, and the latest report of the results of this trial indicates a stratified HR = 0.78; 95% CI, 51.9 to 78.3.⁹⁷

CLEAR trial: Lenvatinib plus pembrolizumab or everolimus versus sunitinib.

The CLEAR phase III trial allocated 1,069 patients with treatment-naïve metastatic RCC to receive lenvatinib (20 mg orally once daily) plus pembrolizumab (200 mg intravenously once every 3 weeks), lenvatinib (18 mg orally once daily) plus everolimus (5 mg orally once daily), or sunitinib (50 mg orally once daily, 4 weeks on and 2 weeks off in each 6 week cycle) in 1:1:1 ratio.¹¹ The primary end point was PFS, which was improved with lenvatinib plus pembrolizumab compared with sunitinib (median, 23.9 v 9.2 months; HR for disease progression or death = 0.39; 95% CI, 0.32 to 0.49; $P < .001$). OS was also significantly improved with lenvatinib plus pembrolizumab than with sunitinib (HR = 0.66; 95% CI, 0.49 to 0.88). 79.2% of the patients in the lenvatinib-plus-pembrolizumab group, 66.1% of the patients in the lenvatinib-plus-everolimus group, and 70.4% of the patients in the sunitinib group were alive at 24 months. The percentage of patients with a confirmed objective response was 71.0% with lenvatinib plus pembrolizumab, 53.5% with lenvatinib plus everolimus, and 36.1% with sunitinib. The percentage of patients with a CR was 16.1% in the lenvatinib-plus-pembrolizumab group, 9.8% in the lenvatinib-plus-everolimus group, and 4.2% in the sunitinib group. Grade 3 or higher AEs of any cause occurred in 82.4% of the patients who received lenvatinib plus pembrolizumab, in 83.1% of the patients who received lenvatinib plus everolimus, and in 71.8% of the patients who received sunitinib. In the lenvatinib-plus-pembrolizumab group, AEs of any

grade led to discontinuation of lenvatinib in 25.6%, pembrolizumab in 28.7%, and both drugs in 13.4%. 68.8% of patients receiving lenvatinib plus pembrolizumab required dose reductions because of the treatment-related AEs compared with 50.3% with sunitinib.

KEYNOTE-426: Pembrolizumab plus axitinib versus sunitinib.

In the phase III KEYNOTE-426 trial, 861 patients with previously untreated advanced RCC were randomly assigned 1:1 to receive pembrolizumab (200 mg, intravenously once every 3 weeks) plus axitinib (5 mg, orally twice daily) or sunitinib (50 mg orally once daily, 4 weeks on 2 weeks off in each 6-week cycle). Primary end points were OS and PFS in the ITT population. After a median follow-up of 30.6 months, pembrolizumab plus axitinib showed continued clinical benefit over sunitinib monotherapy in both OS (median not reached *v* 35.7 months; HR = 0.68) and PFS (15.4 *v* 11.1 months; HR = 0.71). Patients treated with pembrolizumab plus axitinib had a higher ORR (60% *v* 40%) and CR (9% *v* 3%) when compared with sunitinib monotherapy. A lower percentage of patients had primary progressive disease in the pembrolizumab plus axitinib group (11% *v* 17%). 24-month PFS was higher in the pembrolizumab plus axitinib arm (34.3% *v* 22.7%). Serious treatment-related AEs occurred in 28% of the patients treated with pembrolizumab plus axitinib and 16% of the patients treated with sunitinib monotherapy. In the pembrolizumab plus axitinib group, treatment-related AEs led to treatment interruption or discontinuation of pembrolizumab in 44% and 21%, respectively, and axitinib in 62% and 20%, respectively, with discontinuation of both drugs in 7% of the patients. Treatment-related AEs led to interruption in 44% and discontinuation in 12% of sunitinib-treated patients.⁵⁹ In an analysis of HRQoL from the KEYNOTE-426 study, there was a similar HRQoL over 30 weeks for patients receiving pembrolizumab plus axitinib compared with sunitinib.

A 2021 abstract reporting results after a median 42.8 months of follow-up showed the median OS was 45.7 months (95% CI, 43.6 to not reached) with pembrolizumab plus axitinib compared with 40.1 months with sunitinib (95% CI, 34.3 to 44.2; HR = 0.73; 95% CI, 0.60 to 0.88; *P* < .001) and the median PFS was 15.7 months (95% CI, 13.6 to 20.2) with pembrolizumab plus axitinib versus 11.1 months with sunitinib (95% CI, 8.9 to 12.5; HR = 0.68; 95% CI, 0.58 to 0.80; *P* < .0001).⁵⁷

CheckMate 9ER: Nivolumab plus cabozantinib versus sunitinib. In the CheckMate 9ER phase III, randomized, open-label trial, patients with previously untreated advanced ccRCC were randomly assigned to receive either nivolumab (240 mg once every 2 weeks) plus cabozantinib (40 mg once daily) or sunitinib (50 mg once daily 4 weeks on 2 weeks off of each 6-week cycle).¹² The primary end point was PFS by blinded independent central review. Secondary end points included OS, objective response,

and safety. Health-related QoL was an exploratory end point. 651 patients were assigned to receive nivolumab plus cabozantinib (323 patients) or sunitinib (328 patients). At a median follow-up of 18.1 months for OS, the median PFS was 16.6 months (95% CI, 12.5 to 24.9) with nivolumab plus cabozantinib and 8.3 months (95% CI, 7.0 to 9.7) with sunitinib (HR for disease progression or death = 0.51; 95% CI, 0.41 to 0.64; *P* < .001). The probability of OS at 12 months was 85.7% (95% CI, 81.3 to 89.1) with nivolumab plus cabozantinib and 75.6% (95% CI, 70.5 to 80.0) with sunitinib (HR for death = 0.60; 95% CI, 0.40 to 0.89; *P* = .001). An objective response occurred in 55.7% of the patients receiving nivolumab plus cabozantinib and in 27.1% of those receiving sunitinib (*P* < .001). AEs of any cause of grade 3 or higher occurred in 75.3% of the 320 patients receiving nivolumab plus cabozantinib and in 70.6% of the 320 patients receiving sunitinib. Among all treated patients, 56.3% of the patients had a reduction in the dose of cabozantinib, and 51.6% had a reduction in the dose of sunitinib. AEs of any cause led to treatment discontinuation in 19.7% of the patients treated with nivolumab plus cabozantinib (nivolumab only, cabozantinib only, and both drugs in 6.6%, 7.5%, and 5.6% of the patients, respectively) and 16.9% of the sunitinib-treated patients. Better HRQoL was seen with nivolumab plus cabozantinib compared with sunitinib.¹²

Clinical interpretation. For patients with IMDC intermediate- or poor-risk disease who are candidates for treatment with ICIs, a combination regimen including an ICI should be used for the first-line systemic treatment. With five approved combination regimens (Nivo plus Ipi, pembrolizumab plus axitinib, nivolumab plus cabozantinib, avelumab plus axitinib, and pembrolizumab plus lenvatinib), and in the absence of predictive biomarkers or prospective comparative data for these regimens, treatment selection should be based on side effects, comorbid conditions, provider experience, and out-of-pocket treatment costs. A detailed discussion with the patient and shared decision making is the recommended approach. Although QoL was evaluated in several of the studies, differences in QoL methodology and assessment limit the use of the information for decision making among the different treatment regimens.

Recommendation 3.3. Patients with favorable-risk disease who require systemic therapy may be offered an ICI in combination with a VEGFR TKI (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Literature review and analysis. Refer to the literature review for recommendation 3.2.

Clinical interpretation. Among the five approved combinations for the first-line treatment of mRCC (see section 3.2), CheckMate 214 was designed with coprimary end points of ORR, PFS, and OS among IMDC intermediate-

and poor-risk patients. The secondary end points of this trial included ORR, PFS, and OS in the intent-to-treat population that included the IMDC favorable-risk patients. The 42-month follow-up results of this study reported that the median OS was not reached in either arm (Nivo plus Ipi v sunitinib) and the HR for death was 1.19 (95% CI, 0.77 to 1.85) among the exploratory efficacy population of favorable-risk patients. Therefore, the ipilimumab and nivolumab combination is currently approved for patients with IMDC intermediate and poor risk and ICI plus VEGFR TKI combination regimens are recommended for patients with IMDC favorable-risk mRCC, in addition to the intermediate- and poor-risk groups.

Recommendation 3.4. Select patients with metastatic ccRCC receiving systemic therapy in the first-line setting including those with favorable-risk disease or with certain coexisting medical problems may be offered monotherapy with either a VEGFR TKI or an ICI (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Literature review and analysis. Prior studies using the VEGFR TKI sunitinib demonstrated improvement in response and survival outcomes compared with IFN- α , with similar outcomes seen with the VEGFR TKI pazopanib in patients with advanced ccRCC.^{22,92}

In the A031203 CABOSUN trial, cabozantinib, an oral inhibitor of MET, AXL, and VEGFR2, demonstrated an improvement in median PFS compared with sunitinib (8.6 months [95% CI, 6.8 to 14.0] v 5.3 months [95% CI, 3.0 to 8.2]) for cabozantinib (n = 79) versus sunitinib (n = 78; HR = 0.48; 95% CI, 0.31 to 0.74; two-sided $P = .0008$) in patients with previously untreated advanced RCC (intermediate or poor risk by IMDC criteria).⁹³

The phase II KEYNOTE-427 study⁹⁸ evaluated efficacy and safety of single-agent pembrolizumab in treatment-naïve patients with advanced ccRCC (cohort A) and advanced non-ccRCC (cohort B). In cohort A (n = 110), ORR was 36.4% with four (3.6%) CRs and 36 (32.7%) partial responses; disease control rate was 58.2% (95% CI, 48.4 to 67.5). Most patients (68.2%) had a decrease in target lesions, including 30.9% with a reduction $\geq 60\%$. Median duration of response was 18.9 (range, 2.3-37.61) months; 64.1% of responders had a response ≥ 12 months (Kaplan-Meier). Median PFS was 7.1 months (95% CI, 5.6 to 11.0). Median OS was not reached; 12-month and 24-month OS rates were 88.2% and 70.8%, respectively. Durable responses were observed across all IMDC categories. Grade 3-5 treatment-related AEs were reported in 30.0% of patients, of which colitis and diarrhea were most frequent.

Clinical interpretation. Although combination therapy including ICIs is the preferred management strategy, use of VEGFR TKI monotherapy²⁰ or ICI monotherapy such as pembrolizumab⁹⁸ may be considered in select patients.

Risk group and/or coexisting medical issues may preclude use of ICIs or VEGFR TKIs.

Recommendation 3.5. The use of high-dose (HD) IL2 may be considered in the first-line systemic therapy setting for patients with metastatic ccRCC (see *Practical Information*). Attempts to develop criteria to predict those patients most likely to derive benefit from HD-IL2 have been unsuccessful (Type: Evidence based; benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Weak).

Practical Information: The significant toxicity of this regimen must be weighed in relation to the newer immunotherapy regimens that have largely replaced this treatment. The Expert Panel was not able to identify a patient population who should receive this treatment preferentially on the basis of available data. The Expert Panel did agree that HD-IL-2 should be administered in experienced high-volume centers, and that enrollment in clinical trials was preferred.

Literature review and analysis. HD-IL2 can achieve long-lasting remission in a small subset of patients but is associated with substantial toxicity. Attempts to characterize the tumor and patient characteristics that benefit the most have been unsuccessful.⁹⁹

The select trial was designed to prospectively validate predictive biomarkers of response to HD-IL2 in patients with metastatic RCC.⁹⁹ In 120 eligible patients, the ORR was 25% (95% CI, 17.5 to 33.7, $P = .0014$; three CRs, 27 partial responses) with 13 patients (11%) maintaining PFS at 3 years, and the median OS was 42.8 months. Although durable remissions and prolonged survival were seen in a subset of patients, the integrated selection model (eg, clear-cell histology subclassification and CAIX immunohistochemistry staining) was unable to improve the selection criteria.

Clinical interpretation. HD-IL2 can achieve durable remissions and prolonged survival in a small subset of patients but is associated with substantial toxicity. Attempts to use selection criteria to predict those patients most likely to derive benefit from HD-IL2 have been unsuccessful. With newer effective therapies including ICIs and VEGFR TKIs, there is a limited role for HD-IL2 in the management of patients with advanced RCC.

Literature review and analysis. The systematic review identified 17 studies reported in 18 papers¹¹⁻²⁸ reporting on first-line systemic therapy in patients with metastatic ccRCC. A subset of patients with mRCC may have a more indolent disease course such that an initial active surveillance strategy may be considered. In a prospective phase II trial of active surveillance in patients with treatment-naïve, asymptomatic metastatic RCC with a primary end point of time to initiation of systemic therapy, the median time on surveillance until initiation of systemic therapy was 14.9 months (95% CI, 10.6 to 25.0) with more IMDC risk factors and a higher number of metastatic sites associated

with a shorter time on surveillance.^{89,90} Major advances in the first-line management of metastatic ccRCC initially occurred with the development of VEGF-targeted therapy. In 2003, a randomized trial of the anti-VEGF antibody, bevacizumab, versus placebo in patients with progressive metastatic ccRCC and previous therapy with IL-2 (or contraindications to IL-2) demonstrated a significant prolongation of time to progression in patients receiving bevacizumab compared with placebo and heralded the paradigm shift to the use of VEGF-targeted treatments for patients with advanced RCC.⁹¹ Subsequent studies using the VEGFR TKI sunitinib demonstrated improvement in response and survival outcomes compared with IFN- α , with similar outcomes seen with the VEGFR TKI pazopanib leading to a new standard of care for patients with advanced ccRCC.^{22,92} In addition to VEGF TKIs, temsirolimus, an inhibitor of mammalian target of rapamycin kinase demonstrated an improvement in OS compared with IFN- α in patients with poor-prognosis metastatic RCC.²⁸ In the A031203 trial, cabozantinib, an oral inhibitor of MET, AXL, and VEGFR2, demonstrated an improvement in median PFS compared with sunitinib (8.6 months; 95% CI, 6.8 to 14.0) versus 5.3 months (95% CI, 3.0 to 8.2) for cabozantinib versus sunitinib (HR = 0.48; 95% CI, 0.31 to 0.74; two-sided P = .0008) in patients with previously untreated advanced RCC (intermediate or poor risk by IMDC criteria).⁹³ More recently, phase III randomized trials using sunitinib as the de facto comparator arm have led to a new standard of care incorporating ICIs in the treatment of patients with advanced RCC.

CheckMate 214: Nivo plus Ipi versus sunitinib. CheckMate 214 was the pivotal phase III trial investigating the role of dual ICI therapy with an anti-PD-1 (nivolumab) in combination with an anti-CTLA-4 (ipilimumab) in patients with previously untreated advanced renal cell carcinoma. In this study, a total of 1,096 patients were randomly assigned 1:1 to receive nivolumab plus ipilimumab (Nivo plus Ipi) or sunitinib. Nivolumab 3 mg/kg and ipilimumab 1 mg/kg were administered intravenously once every 3 weeks for four cycles followed by nivolumab 3 mg/kg once every 2 weeks. Sunitinib was administered 50 mg orally once daily, alternating 4 weeks receiving treatment and 2 weeks without treatment in each 6-week cycle. Primary end points were OS, PFS, and ORR in the IMDC intermediate-risk/poor-risk population, and secondary end points were OS, PFS, and ORR in the ITT population, and treatment safety. At a minimum study follow-up of 42 months in the primary efficacy population of intermediate-risk/poor-risk patients, Nivo plus Ipi improved OS probability (52% v 39%) and PFS probability (33% v 16%) when compared with sunitinib.¹³ In addition, higher ORR (42% v 26%) and CR rates (10.1% v 1.4%) were seen with Nivo plus Ipi when compared with sunitinib. Among favorable-risk patients, median OS was not reached in either cohort, and the OS probabilities were comparable between the two arms (70% with Nivo plus Ipi v

73% with sunitinib). In the favorable-risk patients, ORR was higher with sunitinib (54% v 29%) but a higher proportion of patients achieved CR with Nivo plus Ipi (12.8% v 5.6%). 42-month PFS was higher in the Nivo plus Ipi arm (32% v 20%) when compared with the sunitinib arm. Grade 3 or higher AEs were seen in 47.3% and 64.1% of the patients treated with Nivo plus Ipi and sunitinib, respectively.⁹⁴

JAVELIN Renal 101: Avelumab plus axitinib versus sunitinib.

The phase III JAVELIN Renal 101 trial enrolled 886 patients with treatment-naïve metastatic ccRCC and randomly assigned them 1:1 to avelumab 10 mg intravenously once every 2 weeks/kg plus axitinib 5 mg orally twice daily or sunitinib 50 mg once daily for 4 weeks every 6 weeks.^{15,95,96} The two independent primary end points were PFS and OS among patients with PD-L1+ tumors. Key secondary end points were OS and PFS in the overall population. Among the 560 patients with PD-L1–positive tumors (63.2%), the median PFS was 13.8 months with avelumab plus axitinib, compared with 7.2 months with sunitinib (HR for disease progression or death = 0.61; 95% CI, 0.47 to 0.79; P < .001); in the overall population, the median PFS was 13.8 months, compared with 8.4 months (HR = 0.69; 95% CI, 0.56 to 0.84; P < .001). Among the patients with PD-L1–positive tumors, the ORR was 55.2% with avelumab plus axitinib and 25.5% with sunitinib; at a median follow-up for OS of 11.6 months and 10.7 months in the two groups, 37 patients and 44 patients had died, respectively. Among patients who received axitinib in the combination group, 42.2% had at least one reduction in the dose of axitinib and 10.8% had at least one escalation in the dose of axitinib. Of the patients who received sunitinib, 42.6% had at least one dose reduction. AEs of grade 3 or higher during treatment occurred in 71.2% in the avelumab plus axitinib group and 71.5% in the sunitinib group. AEs that occurred during treatment led to discontinuation of both avelumab and axitinib in 7.6% who received the combination and led to discontinuation of sunitinib in 13.4% who received sunitinib. Stratified analyses demonstrated relatively consistent benefit of avelumab and axitinib, compared with sunitinib, across strata of IMDC risk groups. These effects were generally statistically significant for PFS and ORR (apart from PFS among patients with IMDC good-risk disease), whereas the effect was statistically significant for OS only among those with IMDC intermediate-/poor- and poor-risk disease.

CLEAR trial: Lenvatinib plus pembrolizumab or everolimus versus sunitinib.

The CLEAR phase III trial randomly assigned 1,069 patients with treatment-naïve metastatic RCC to receive lenvatinib (20 mg orally once daily) plus pembrolizumab (200 mg intravenously once every 3 weeks), lenvatinib (18 mg orally once daily) plus everolimus (5 mg orally once daily), or sunitinib (50 mg orally once daily, alternating 4 weeks receiving treatment and 2 weeks without treatment) in 1:1:1 ratio.¹¹ The primary end point was PFS, which was met and was longer with lenvatinib

plus pembrolizumab than with sunitinib (median, 23.9 v 9.2 months; HR for disease progression or death = 0.39; 95% CI, 0.32 to 0.49; $P < .001$). OS was also significantly longer with lenvatinib plus pembrolizumab than with sunitinib (HR = 0.66; 95% CI, 0.49 to 0.88). 79.2% of the patients in the lenvatinib-plus-pembrolizumab group, 66.1% of the patients in the lenvatinib-plus-everolimus group, and 70.4% of the patients in the sunitinib group were alive at 24 months. The percentage of patients with a confirmed objective response was 71.0% with lenvatinib plus pembrolizumab, 53.5% with lenvatinib plus everolimus, and 36.1% with sunitinib. The percentage of patients with a CR was 16.1% in the lenvatinib-plus-pembrolizumab group, 9.8% in the lenvatinib-plus-everolimus group, and 4.2% in the sunitinib group. Grade 3 or higher AEs of any cause occurred in 82.4% of the patients who received lenvatinib plus pembrolizumab, in 83.1% of the patients who received lenvatinib plus everolimus, and in 71.8% of the patients who received sunitinib. In the lenvatinib-plus-pembrolizumab group, AEs of any grade led to discontinuation of lenvatinib in 25.6%, pembrolizumab in 28.7%, and both drugs in 13.4%. 68.8% of patients receiving lenvatinib plus pembrolizumab required dose reductions because of the treatment-related AEs compared with 50.3% with sunitinib.

KEYNOTE-426: Pembrolizumab plus axitinib versus sunitinib.

In the phase III KEYNOTE-426 trial, 861 patients with previously untreated advanced RCC were randomly assigned 1:1 to receive pembrolizumab (200 mg, intravenously once every 3 weeks) plus axitinib (5 mg, orally twice daily) or sunitinib (50 mg orally once daily, alternating 4 weeks receiving treatment and 2 weeks without treatment in each 6 weeks cycle). Primary end points were OS and PFS in the ITT population. After a median follow-up of 30.6 months, pembrolizumab plus axitinib showed continued clinical benefit over sunitinib monotherapy in both OS (HR = 0.68; median not reached v 35.7 months) and PFS (HR = 0.71; 15.4 v 11.1 months). Patients treated with pembrolizumab plus axitinib had a higher ORR (60% v 40%) and CR (9% v 3%) when compared with sunitinib monotherapy. A lower percentage of patients had primary progressive disease in the pembrolizumab plus axitinib group (11% v 17%). Serious treatment-related AEs occurred in 28% of the patients treated with pembrolizumab plus axitinib and 16% of the patients treated with sunitinib monotherapy. Twenty-four-month PFS was higher in the pembrolizumab plus axitinib arm (34.3% v 22.7%).

Although combination therapy including ICIs is the preferred management strategy, use of VEGFR TKI monotherapy including cabozantinib, pazopanib, sunitinib, and axitinib, or ICI monotherapy such as pembrolizumab may be considered in select patients on the basis of risk group and/or coexisting medical issues that may preclude use of ICIs or VEGFR TKIs.

CheckMate 9ER: Nivolumab plus cabozantinib versus sunitinib. In the CheckMate 9ER phase III, randomized, open-label trial, patients with previously untreated advanced ccRCC were randomly assigned to receive either nivolumab (240 mg once every 2 weeks) plus cabozantinib (40 mg once daily) or sunitinib (50 mg once daily for 4 weeks of each 6-week cycle).¹² The primary end point was PFS by blinded independent central review. Secondary end points included OS, objective response, and safety, and HRQoL was an exploratory end point. Six hundred fifty-one patients were assigned to receive nivolumab plus cabozantinib (323 patients) or sunitinib (328 patients). At a median follow-up of 18.1 months for OS, the median PFS was 16.6 months (95% CI, 12.5 to 24.9) with nivolumab plus cabozantinib and 8.3 months (95% CI, 7.0 to 9.7) with sunitinib (HR for disease progression or death = 0.51; 95% CI, 0.41 to 0.64; $P < .001$). The probability of OS at 12 months was 85.7% (95% CI, 81.3 to 89.1) with nivolumab plus cabozantinib and 75.6% (95% CI, 70.5 to 80.0) with sunitinib (HR for death = 0.60; 98.89% CI, 0.40 to 0.89; $P = .001$). An objective response occurred in 55.7% of the patients receiving nivolumab plus cabozantinib and in 27.1% of those receiving sunitinib ($P < .001$). AEs of any cause of grade 3 or higher occurred in 75.3% of the 320 patients receiving nivolumab plus cabozantinib and in 70.6% of the 320 patients receiving sunitinib. Better HRQoL was seen with nivolumab plus cabozantinib compared with sunitinib.

Clinical interpretation. The preferred management for first-line systemic therapy includes a combination with two ICIs or an ICI plus a VEGFR TKI. With five approved combination regimens (Nivo plus Ipi, pembrolizumab plus axitinib, nivolumab plus cabozantinib, avelumab plus axitinib, and pembrolizumab plus lenvatinib) the choice of treatment should be based on QoL, side effects, comorbid conditions, and out-of-pocket treatment costs. Combination therapy is preferred; however, select patients may be considered for treatment with monotherapy, ie, an ICI or VEGFR TKI. In addition, an initial active surveillance strategy may be considered for select patients.

Clinical Question 4

What is the optimal second- or later-line systemic treatment for metastatic clear cell renal cell carcinoma?

Recommendation 4.1. Nivolumab or cabozantinib should be offered to patients who progressed on a VEGFR TKI alone (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Literature review and analysis. A literature review was performed identifying randomized trials evaluating treatment in patients with ccRCC after prior VEGFR TKI. Of these studies, the following had OS benefit:

METEOR: Allocated 658 patients who had at least one prior TKI to cabozantinib or everolimus. OS favored cabozantinib (20.4 months) over everolimus (17.1 months; $P = .0002$).³³

INTORSECT: Compared temsirolimus against sorafenib in patients (N = 512) who had progressed on first-line sunitinib. PFS was similar (4.3 months v 3.9 months, respectively), whereas OS favored sorafenib (16.6 months versus 12.3 months, respectively; $P = .01$).⁴²

CheckMate 025: Patients who had received one or two lines of VEGFR TKI were randomly assigned to nivolumab or everolimus (N = 821). OS favored the nivolumab arm (25.8 months v 19.7 months, HR = 0.73). PFS and response rates also favored the nivolumab arm.³⁰

METEOR and CheckMate 025 resulted in some of the strongest OS data in second-line therapy after VEGFR TKI. Selection between cabozantinib and nivolumab must naturally include an individualized assessment relative to pace of disease progression, presence of disease-related symptoms, patient comorbidities, and other factors such as financial toxicity and travel concerns.

Additional studies have shown PFS benefit for other treatments that could also be considered, including axitinib and lenvatinib plus everolimus.

AXIS: Patients may have received one prior therapy with either sunitinib, bevacizumab plus interferon, temsirolimus, or cytokine, and were randomly assigned to axitinib or sorafenib. OS was not statistically significant (20.1 months v 19.2 months, respectively, $P = .3744$) despite a benefit in PFS favoring axitinib (N = 723).⁴⁴

Motzer et al⁴⁰ reported on a 3-arm study comparing lenvatinib versus everolimus versus lenvatinib plus everolimus in 153 subjects. The lenvatinib and lenvatinib plus everolimus arms had superior PFS compared with everolimus. This study was not powered for OS, but the combination was approved by the US Food and Drug Administration (FDA) in 2015.⁴⁰

The available data suggest that treating with a different VEGFR TKI after a prior TKI can yield responses and clinical benefit. However, given the potential for durable responses and milder toxicity of nivolumab, this may be a more optimal choice for appropriate patients after an initial VEGFR TKI. Unfortunately, in many aspects of the kidney cancer treatment spectrum, we lack direct head-to-head comparisons to make unequivocal recommendations (eg, there is no direct comparison of nivolumab v cabozantinib in the second line).

Clinical interpretation. Most patients with mRCC will experience disease progression on first-line therapy but will generally be in good performance status to consider second-line therapies or beyond. The choice of therapy will depend on which agents were used in first-line therapy, magnitude, and duration of response as well as patient preferences and comorbidities. Clinical trial participation is strongly suggested. Patients can be grouped into those that had TKI alone, immunotherapy alone (single agent or combination immunotherapy), or combination TKI with immunotherapy.

Recommendation 4.2. Patients progressing on combination immunotherapy (eg nivolumab and ipilimumab) should be offered a VEGFR TKI (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Literature review and analysis. There are few randomized studies evaluating second-line therapy after ICI failure. As such, no standard has been set for patients in this situation. Given that the Nivo plus Ipi combination has used agents from both immunotherapy groups (PD-1/PD-L1 inhibitors and anti-CTLA-4), the most logical option is to recommend a VEGFR TKI. On the basis of strength of data from front-line therapy studies, cabozantinib and lenvatinib plus everolimus are reasonable options. There are subsets of patients in a few studies that had received immunotherapy alone in first-line treatment, who then received a TKI and ICI combination such as lenvatinib plus pembrolizumab, but no randomized trials have been reported to date that prove that continuation of immunotherapy is beneficial over VEGFR TKI monotherapy.

Clinical interpretation. Considering that the combination of nivolumab and ipilimumab includes a drug from each of the main immunotherapy subclasses (anti-PD-1/PD-L1 and anti-CTLA-4), it is unlikely there will be a benefit to switching to a different immunotherapy drug of a similar class. Thus, switching to a VEGFR TKI (alone or in combination) would be preferred in this setting, and ongoing trials are testing this approach.

Recommendation 4.3. Patients who progress after initial therapy combining VEGFR TKI with an ICI may be offered an alternate VEGFR TKI as a single agent (Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Literature review and analysis. Since most of the VEGFR TKI and ICI combinations are relatively new, there are not any randomized trials that establish an absolute standard of care except for the TIVO-3 trial,¹⁰⁰ which included a subset of patients who had been treated with both a VEGFR TKI and an ICI (sequentially or in combination). In this study, 350 patients were randomly assigned to either tivozanib or sorafenib. Tivozanib showed improved PFS over sorafenib (5.6 months v 3.9 months; $P = .016$). Although OS was not statistically different (tivozanib 16.2 months v sorafenib 19.6 months; HR = 0.97), tivozanib received FDA approval for use in refractory or relapsed kidney cancer.¹⁰⁰

Clinical interpretation. This is one of the few trials in the late-stage line of therapy that has shown some clinical benefit against an established agent. Although there is no survival benefit, in the absence of agents with better outcomes, tivozanib is a reasonable option.

Recommendation 4.4. For patients on immunotherapy who experience limited disease progression (eg, one site of progression), local therapy (radiation, thermal ablation, and

excision) may be offered, and immunotherapy may be continued (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Weak).

Literature review and analysis. There are no randomized trials evaluating this approach. Several studies have reported primarily on the safety of concurrent stereotactic radiation with systemic therapy. Disease control is reported for a large subset of these patients.

Clinical interpretation. Although there is an absence of prospective randomized trials evaluating this approach, the committee agreed this was a practical approach with potential for benefiting patients. Case series reporting an abscopal effect in patients treated with radiation or cryotherapy primarily for limited progressive disease provide anecdotal support for this option. Additional data from controlled clinical studies are needed.

Clinical Question 5

What is the optimal application of metastasis-directed therapy for metastatic clear cell renal cell carcinoma?

Recommendation 5.1. For patients with low-volume mRCC, definitive metastasis-directed therapies may be offered and include surgical resection (metastasectomy), ablative measures, or radiotherapy (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 5.2. For patients undergoing complete metastasectomy, subsequent TKIs are not routinely recommended (Type: Evidence based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Literature review and analysis. The majority of data to date have been retrospective in nature, with recent systematic reviews for both metastasectomy¹⁰¹ and stereotactic ablative body radiotherapy (SABR)⁸ in advanced RCC identifying only retrospective series. Since these systematic reviews, there have been three published prospective trials. For reference, SABR is defined¹⁰² as a treatment that is accurate and precise, and has a HD (ie, > typically 8 Gy/fraction, up to approximately 5 fractions) with sharp dose fall-off that minimizes toxicity on nearby organs.

The RESORT study addressed the addition of TKIs to total definitive metastasectomy. This study allocated 33 patients to total metastasectomy, compared with 36 patients allocated to total metastasectomy followed by sorafenib. The median follow-up was 38 months. The 2-year and 3-year disease-free survival probability was 59% versus 49%, and 50% versus 41%, respectively, in the observation arm versus the sorafenib arm.⁵¹ The authors concluded no benefit to the addition of sorafenib to metastasectomy. This is similar to the findings of the ECOG-ACRIN E2810 study (reported in abstract form), which found no benefit to the addition of pazopanib to complete metastasectomy.¹⁰³

For oligoprogressive disease, in which limited sites of disease are progressing despite otherwise effective systemic therapy, SABR has been prospectively evaluated.¹⁰⁴ This prospective single-arm multicenter study enrolled a total of 37 patients. With TKI therapy, SABR to oligoprogressive metastases is associated with 93% local control and 92% OS at 1 year. The median extended PFS was 9.3 months, and the median time to switching of systemic therapy was 12.6 months. The authors concluded that SABR is a viable strategy in oligoprogressive disease that can significantly delay the need to change to the next line of drug therapy.

A further phase II single-arm trial of 30 patients with oligometastatic (ie, five or fewer lesions) RCC undergoing metastasis-directed therapy with SABR¹⁰⁵ has been published. At a median follow-up of 17.5 months, 1-year PFS was 64%, local control rate was 97%, and systemic therapy-free survival was 82%. The authors concluded that in selected patients, SABR could be used in lieu of systemic therapy with minimal associated toxicity.

For patients receiving concurrent SABR and ICIs, several studies indicate the safety of this combination. The RAP-PORT study¹⁰⁶ (ClinicalTrials.gov identifier: [NCT02855203](https://clinicaltrials.gov/ct2/show/study?term=NCT02855203)) was a multi-institutional single-arm, phase I/II study of SABR and short-course pembrolizumab (eight cycles) in oligometastatic RCC. There were no grade 4+ events and 13% grade 3 events. The 2-year PFS and local control were 45% and 92%, respectively. The NIVES study¹⁰⁷ evaluated the addition of SABR with nivolumab; 26% experienced grade 3-4 treatment-related AEs and 7% discontinued therapy. In a meta-analysis¹⁰⁸ of radiotherapy and ICIs, there were similar grade 3-4 toxicities in ICIs plus radiotherapy (17.8%; 95% CI, 12.0 to 24.5) compared with ICIs alone (22.4%; 95% CI, 18.4 to 26.6). In a meta-analysis¹⁰⁹ of ICIs and TKIs, the addition of TKIs to radiotherapy-based treatment did not improve OS (HR = 1.02; 95% CI, 0.90 to 1.15, $P = .76$) but increased grade 3+ toxicity (relative risk = 1.18; 95% CI, 1.06 to 1.33; $P = .009$).

Clinical interpretation. Both metastasectomy and SABR are currently considered treatment options in the National Comprehensive Cancer Network¹¹⁰ and the European Association of Urology management guidelines¹¹¹ for oligometastatic RCC, with the goal of local palliation, or to ablate all sites of macroscopic disease in patients with favorable disease factors. There have not been randomized controlled trials comparing these metastasectomy versus SABR, or whether the omission of these metastasis-directed therapies can be afforded with modern systemic therapies.

Despite this, there is a paucity of prospective clinical trials for these modalities. Surgery is the historical default option for metastasis-directed therapy. There are no prospective trials evaluating surgery as an individual modality; however, a randomized study did not show a benefit of adding sorafenib to metastasectomy alone.⁵¹ As an individual

modality, two surgical metastasectomy systematic reviews of retrospective data have been published, and the authors of these works conclude that surgical metastasectomy may improve survival for select patients.^{101,112}

Surgery may not be an option because of oligoprogression (thus necessitating multiple surgeries), potential surgical morbidity, and patient inoperability (eg, because of comorbidities and advanced age). With the advent of high-dose per fraction, SABR has been gaining popularity as an alternative therapeutic modality.¹¹³

Optimal selection of metastasis-directed therapy is dependent on both patient and treatment factors. Surgery is preferred when histologic confirmation of metastasis is required, in younger and fitter patients, and in the situation of solitary lung and adrenal metastasis. For bone and brain metastases, radiotherapy is often a preferred approach because of the potential morbidity of complete macroscopic excision. Additionally, in patients with low-volume multiple metastases, or low-volume multiorgan metastases, SABR may be a less morbid approach than total metastasectomy. SABR is preferred for patients with oligoprogressive disease because of low morbidity and limited need to interrupt systemic therapy. Current data suggest that concurrent radiotherapy plus ICIs are safe, and that concurrent TKIs should be used with caution.

Clinical Question 6

What considerations should be applied to treatment of special subsets of metastatic clear cell renal cell carcinoma (eg, bone metastases, brain metastases, and sarcomatoid carcinomas)?

Recommendation 6.1.1. Patients with symptomatic bone metastases from metastatic ccRCC should receive bone-directed radiation (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 6.1.2. Patients with bone metastases from metastatic ccRCC should be offered a bone resorption inhibitor (either bisphosphonate or receptor activator of nuclear factor kappa-B ligand [RANKL] inhibitor) when clinical concern for fracture or skeletal-related events (SREs) is present (Type: Consensus based, benefits outweigh harms; Evidence quality: Moderate; Strength of recommendation: Strong).

Recommendation 6.1.3. No recommendation regarding optimal systemic treatment for metastatic ccRCC patients with bone metastasis can be made; however, it is our expert opinion that cabozantinib-containing regimens may be preferred (Type: Consensus based, benefits outweigh harms; Evidence quality: Low; Strength of recommendation: Moderate).

Literature review and analysis. The systematic review identified two phase III trials in patients with VEGF-refractory mRCC, with subgroup analyses on bone metastases. In the

first trial, METEOR, 77 patients with bone metastases treated with cabozantinib were compared with 65 patients with bone metastases treated with everolimus, with improvement in both median OS (20.1 months v 12.1 months, HR = 0.54; 95% CI, 0.34 to 0.84; $P < .05$, P value NR) and median PFS (7.4 months v 2.7 months, HR = 0.33; 95% CI, 0.21 to 0.51; $P < .05$, P value NR) for patients treated with cabozantinib. The second trial (CheckMate 025) compared 76 patients with bone metastases treated with nivolumab to 70 patients with bone metastases treated with everolimus, with nonstatistically significant trend toward improving median OS (18.5 months v 13.8 months, HR = 0.72; 95% CI, 0.47 to 1.09; $P > .05$). In the METEOR trial, 73% of patients with bone metastases treated with cabozantinib had a grade 3-5 AE, compared with 51% of patients with bone metastases treated with everolimus. In the CheckMate 025 trial, grade 3-5 AE event rates were similar (17% v 19%) for patients with bone metastases.

Within first-line phase III trials, both CheckMate 214¹³ and CheckMate 9ER¹² reported clinical outcomes in the subset of patients with bone metastases. In CheckMate 214, 84 patients with bone metastases and IMDC intermediate/poor risk (20%) were treated with Nivo plus Ipi and 89 patients with bone metastases and IMDC intermediate/poor risk (21%) were treated with sunitinib. OS trended toward improvement (HR = 0.71; 95% CI, 0.47 to 1.08) for patients treated with ipilimumab-nivolumab. The phase III first-line trial CheckMate 9ER also reported clinical outcomes from a subset of patients with bone metastases. Seventy-eight patients (24%) with bone metastases were treated with cabozantinib-nivolumab and 72 patients (22%) with bone metastases were treated with sunitinib, with improved PFS (HR = 0.34; 95% CI, 0.22 to 0.55) and OS (HR = 0.54; 95% CI, 0.32 to 0.92) for patients treated with cabozantinib-nivolumab. Patients with bone metastases were included in other randomized studies including KEYNOTE-426, JAVELIN Renal 101, and the CLEAR trials without specific report of outcomes in this subset.

Clinical interpretation. Patients with bone metastases are usually included in prospective clinical trials, and the presence of bone metastases has intermittently been used as a stratification factor during treatment randomization, as bone metastases portend worse prognoses. Subset analyses of patients with bone metastases are variably reported, depending on the trial. There are retrospective single-center and multicenter studies for evaluation of oligometastatic disease with bone metastases treated with radiation.¹¹⁴⁻¹¹⁶ These studies have posited benefits of symptomatic control and delay in systemic therapy initiation.

Patients with bone metastases are at higher risk of SREs (defined as fracture, spinal cord compression, necessity for radiation, or surgery to bone). Two classes of bone-resorptive agents of bisphosphonates or RANKL inhibitors are widely used in solid malignancies with bone metastases to prevent or delay time to SREs. The definitive trial randomly assigning

patients with solid tumors to either zoledronic acid or placebo delayed time to first SRE across tumor types. The cohort with bone mRCC similarly benefited: the rate of SREs, annual incidence of SREs, and time to progression of bone metastases were all improved for patients with mRCC bone metastases treated with zoledronic acid versus placebo.¹¹⁷ In addition, patients with mRCC randomly assigned to first-line zoledronic acid with everolimus versus everolimus alone also delayed time to first SRE.¹¹⁸ Other retrospective series have demonstrated safety of concurrent treatment with zoledronic acid and VEGF TKIs.^{119,120} In recent years, denosumab as a RANKL inhibitor has been compared with zoledronic acid demonstrating at least noninferiority or even superiority in a meta-analysis^{117,121}; therefore, our recommendation encompasses both agents. In eligible patients and with appropriate monitoring for toxicities, particularly renal insufficiency, a bisphosphonate or RANKL inhibitor should be offered.

Prospective trials specifically studying patients with mRCC and bone metastases will further improve understanding of optimal treatments. A prospective trial, RADICAL (Clinical-Trials.gov identifier: [NCT04071223](#)), is currently enrolling patients with mRCC with bone metastases and evaluating treatment with cabozantinib and radium-223 dichloride. When deciding upon systemic therapy for patients with bone metastasis, regimens containing cabozantinib should be considered, given the currently available data.

Recommendation 6.2.1. Patients with brain metastases from metastatic ccRCC should receive brain-directed local therapy with radiation therapy and/or surgery (Type: Consensus based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Recommendation 6.2.2. No recommendation regarding optimal systemic therapy for patients with metastatic ccRCC and brain metastases can be made (Type: Consensus based, benefits to harms ratio unknown; Evidence quality: NA; Strength of recommendation: Strong).

Literature review and analysis. The literature review revealed two prospective studies of patients with metastatic RCC to the brain. In the GETUG-AFU 26 NIVOREN study (Clinical-Trials.gov identifier: [NCT03013335](#)), patients with untreated brain metastases (cohort A, n = 39) and previously treated brain metastases (cohort B, n = 34) were treated with nivolumab monotherapy 240 mg intravenously once every 2 weeks. The ORR by modified RECIST v1.1 in cohort A was 12% with no responses seen in lesions > 1 cm and with 13% of enrolled patients being unevaluable because of rapid clinical progression and death before the first evaluation. The median intracranial PFS was 2.7 months in cohort A and 4.8 months in cohort B, and most patients in cohort A required subsequent brain metastasis-directed local therapy (72%). The intracranial response rate of 12% was lower than the ORR in extracranial tumors (21%). Prior focal brain therapy (cohort B)

decreased the risk of intracranial progression after adjustment for baseline characteristics (HR = 0.49; 95% CI, 0.26 to 0.92). There were no surprising safety signals.

The multicohort phase IIIb/IV CheckMate 920 study treated 28 patients with brain metastases with first-line Nivo plus Ipi with a primary end point of incidence of grade ≥ 3 immune-mediated AEs. Patients were required to be asymptomatic from their brain metastases with no systemic corticosteroid treatment or radiation treatment within 14 days before study treatment, and most patients (89%) had received prior therapy for brain metastasis. This study showed an acceptable rate of immune-related AEs (54% of patients had grade 3-4 treatment-related AEs) and an overall response rate of 32%; intracranial response rate was NR with 25% of patients demonstrating intracranial progression. The median PFS on this study was 9.0 months.¹²²

Clinical interpretation. Patients with brain metastases are frequently excluded from clinical trials, including complete exclusion of all patients with history of brain metastases from CheckMate 214, CheckMate 9ER, and CheckMate-025, and exclusion of patients with active or symptomatic brain metastases in KEYNOTE-426, CLEAR, and Javelin Renal 101. Therefore, most data available are from expanded access programs, retrospective studies, and a small number of prospective single-arm studies. Prospective data are lacking to support recommend of any individual systemic treatment, and overall efficacy of systemic treatment is low for control of metastatic RCC in patients with brain metastases. Given the lack of demonstrated intracranial disease control in prospective trials for patients with untreated brain metastases, local therapy should be the standard of care with radiotherapy and/or surgery with decision making individualized on the basis of the pattern of intracranial metastases. A recent ASCO-SNO-ASTRO guideline¹²³ on management of brain metastases includes further discussion on decision making for local therapy.

In regard to other data on systemic therapy, one prospective study of sunitinib in 16 patients with untreated brain metastases has been reported with no objective responses seen in the brain metastases.¹²⁴ However, a larger expanded-access study by Gore et al¹²⁵ reported on 321 patients with baseline brain metastases treated with sunitinib. Twelve percent of 213 evaluable patients had an objective response, and the median PFS was 5.6 months and the median OS was 9.2 months.¹²⁵

Several large retrospective cohort studies suggest antitumor activity in RCC brain metastases with newer agents, including cabozantinib and Nivo plus Ipi. A large retrospective cohort study described the activity of cabozantinib in patients with both untreated and treated brain metastases from 15 institutions. Cohort A (n = 33) included patients with progressing brain metastases who did not receive local brain therapy within the past 2 months and

cohort B (n = 55) included patients with stable or progressing brain metastases who received concomitant brain-directed therapy. The intracranial response rate in cohort A was 55% with a median time to treatment failure of 8.9 months (PFS NR).¹²⁶ Interpretation of this study is limited by the retrospective analysis, the lack of information on steroid use, site assessment of modified RECIST response, and the fact that rapid progressors were excluded because of the need for at least one follow-up imaging assessment after starting cabozantinib. In the CABOREAL French expanded-access program of cabozantinib, 64 patients with baseline brain metastases were followed with a median time on treatment of 6.7 months and a median OS of 11.3 months.¹²⁷ Prospective validation of the activity of cabozantinib in patients with brain metastases is awaited in the CABRAMET study.

Recommendation 6.3. Patients with metastatic ccRCC with sarcomatoid features should receive ICI-based combination first-line treatment (Nivo plus Ipi, or alternatively, an ICI plus a TKI; Type: Evidence based, benefits outweigh harms; Evidence quality: High; Strength of recommendation: Strong).

Literature review and analysis. The systematic review identified one post hoc exploratory analysis of a randomized controlled trial supporting the use of immunotherapy-based combination therapy when compared with sunitinib in patients with ccRCC with a sarcomatoid component. The CheckMate 214 study randomly assigned patients with advanced or metastatic ccRCC to combination ipilimumab and nivolumab or sunitinib. Patients with intermediate- or poor-risk RCC with sarcomatoid features (n = 139, representing 16.4% of the overall trial intermediate-/poor-risk population) who received ipilimumab and nivolumab had improved outcomes compared with those allocated to sunitinib. At a median study follow-up of 42 months, the OS was not reached (95% CI, 25.2 to not estimable) in the ipilimumab and nivolumab arm compared with 14.2 months in the sunitinib arm (9.3 to 22.9) with a HR for death = 0.45 (95% CI, 0.3 to 0.7; $P = .0004$). Median PFS was also longer in the ipilimumab and nivolumab arm compared with sunitinib (26.5 v 5.1 months, HR = 0.54; 95% CI, 0.3 to 0.9, $P = .0093$). In addition, patients who received ipilimumab and nivolumab had higher ORRs than those who received sunitinib (60.8% v 23.1%; $P < .0001$) with a CR rate of 18.9% and 3.1%, respectively.

The Javelin Renal 101 study of axitinib plus avelumab compared with sunitinib also reported a subset analysis of patients with sarcomatoid histology.⁹⁷ This study showed that avelumab plus axitinib was associated with higher ORR (47% v 21%) and longer median PFS (7.0 v 4.0 months, HR = 0.57; 95% CI, 0.33 to 1.00) compared with sunitinib. However, this study has yet to demonstrate an OS benefit in the overall study population or the subset of sarcomatoid patients. The review also identified one single-arm phase II

study of gemcitabine and axitinib in patients with sarcomatoid features that demonstrated an ORR of 56% in 25 patients with a median PFS of 4.2 months and a median OS of 8.4 months.¹²⁸

Clinical interpretation. Any histologic subtype of RCC can contain sarcomatoid features, which have historically been associated with an aggressive phenotype, and presence of sarcomatoid features is an independent adverse prognostic factor. Combination ipilimumab and nivolumab is highly active and improves outcomes compared with sunitinib in patients with sarcomatoid features. Several combination studies of ICI plus VEGF TKI trials have also been presented in abstract form for the subset of patients with sarcomatoid features. In the KEYNOTE-426 study, combination therapy with pembrolizumab plus axitinib resulted in higher response rates (58.8% v 31.5%) compared with sunitinib. Patients on pembrolizumab plus axitinib also had higher median PFS (NR v 8.4 months, HR = 0.54; 95% CI, 0.29 to 1.00) and higher OS (median NR either arm, HR = 0.58; 95% CI, 0.21 to 1.59) compared with patients on sunitinib.¹²⁹ Similarly, the subset of patients with sarcomatoid differentiation in the CheckMate 9ER trial who received combination nivolumab and cabozantinib had higher ORR (55.9% v 22.0%), median PFS (10.9 v 4.2 months, HR = 0.39; 95% CI, 0.22 to 0.70), and median OS (NR v 19.7 months, HR = 0.36; 95% CI, 0.16 to 0.82) compared with those who received sunitinib.¹³⁰ Therefore, first-line treatment of patients with sarcomatoid features should include an ICI combination regimen, given the consistent better outcomes compared with sunitinib alone. Choice of the ICI partner agent (dual immune checkpoint inhibition or VEGFR TKI) should be individualized on the basis of individual patient or provider assessment of preferences, risks, and benefits.

DISCUSSION

The goal of this guideline is to provide a high-level assessment and expert interpretation of treatment recommendations for patients with metastatic ccRCC. In just over a decade, this malignancy has gone from orphan disease status to mainstream, with multiple treatment options available, and the need for guidelines.

As ccRCC has emerged as a disease in which first-in-class drugs are tested against proven biological drivers, we are poised to see additional agents emerging in coming days. For example, the hypoxia inducible factor 2 inhibitor, belzutifan, recently earned FDA approval for the medical control of primary von Hippel-Lindau syndrome-associated renal cell carcinomas.¹³¹ This agent is actively being assessed in the metastatic setting. This and other agents will need to be critically evaluated and applied optimally to provide the best outcomes for patients. The principles applied to arrive at these recommendations will serve us well as we pave the way for future advances that continue to

extend survival and create opportunities for cure for patients living with metastatic ccRCC.

Continued work is needed to define predictive and prognostic biomarkers to further tailor treatments to tumors to increase efficacy while decreasing toxicities (physiologic, psychologic, and financial). This is an especially important consideration, given the rise of doublet standard-of-care treatments and the approval of adjuvant therapy for high-risk RCC after nephrectomy. Clinical trials remain the preferred option for many patients with RCC and are the only way to answer the important questions outlined in this guideline. Therefore, access and equity must be a top priority.

PATIENT AND CLINICIAN COMMUNICATION

The following items were identified by the two patient representatives on the panel as additional factors that both patients and clinicians should consider and address throughout the emotional and medical journeys of someone facing metastatic renal clear cell carcinoma:

- This patient group faces daunting median survival odds. The first-line treatments outlined in these recommendations will likely provide a survival benefit to many patients. However, when disease is no longer being controlled, patients may benefit by focusing on QoL and time spent with loved ones. Palliative care can provide important benefits for this patient group, even when active treatment plans are in play.
- This guideline represents diligent and exhaustive work by leading peers to address current issues and controversies by finding the best evidence available. The goal has been to assist clinicians facing enormous workloads and little time with insights on the latest data to help inform their treatment decisions.
- This patient community has substantially benefitted from the many developments addressed in this guideline. However, the current pace of developments and change places a burden on clinicians and patients to recognize the rapidly changing environment and the need to stay abreast. Second opinions with kidney cancer content experts should become an accepted practice for patients, and case conferences or tumor boards should focus on collaborating on difficult cases to the greatest extent possible.
- The level of uncertainty associated with treatment is high, and this would indicate a substantially greater need for shared decision making between clinicians and patients. Additionally, the care of patients with mRCC requires input from multidisciplinary team members.
- Patients and caregivers should be encouraged to reach out to family members, friends, and other patients to ensure they can avoid the loneliness, depression, and poor decision making that frequently occur by those attempting to go it alone.

- There is great need to improve personalized treatment for patients with metastatic ccRCC. Substantial progress requires better establishing patient criteria for each of the recommended treatments discussed before. Identifying those who have the most efficacious outcomes for each treatment can potentially improve survival, reduce patient cost, and increase medical institution effectiveness. Understanding nonresponders and those patients experiencing negative results is critically important.

For recommendations and strategies to optimize patient-clinician communication, see Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.¹³²

HEALTH DISPARITIES

Although ASCO clinical practice guidelines represent expert recommendations on the best practices in disease management to provide the highest level of cancer care, it is important to note that many patients have limited access to medical care. Racial and ethnic disparities in health care contribute significantly to this problem in the United States. Patients with cancer who are members of racial and/or ethnic minorities suffer disproportionately from comorbidities, experience more substantial obstacles to receiving care, are more likely to be uninsured, and are at greater risk of receiving care of poor quality than other Americans.¹³³⁻¹³⁶ Many other patients lack access to care because of their geographic location and distance from appropriate treatment facilities. Awareness of these disparities in access to care should be considered in the context of this clinical practice guideline, and health care providers should strive to deliver the highest level of cancer care to these vulnerable populations.

mRCC is not immune from these biases in health care. Recent interrogation of the NCDB disclosed disparities in the distribution of systemic therapy and offering of CN.^{72,76} This study found that females are less likely to receive systemic therapy (odds ratio [OR] = 0.91, $P < .01$) and a higher proportion receive no treatment (OR = 1.14, $P < .01$). Non-Hispanic African Americans and Hispanics were less likely to receive CN (OR = 0.75, $P < .01$ and OR = 0.86, $P = .01$, respectively), and African American patients had lower odds of receiving systemic therapy (OR = 0.85, $P < .01$) and increased odds of no treatment (OR = 1.41, $P < .01$). The International Marker Consortium for Renal Cancer (INMARC) database was analyzed to specifically evaluate associations of African American race and survival, which showed that 5-year OS was worse for African Americans with stage IV disease (23% v 44%, $P = .009$) and 5-year cancer-specific survival was also worse for African Americans (30% v 49%, $P = .007$).¹³⁷ By contrast, studies examining Asian Americans with renal cell carcinoma identified no difference in tumor characteristics or survival outcomes when compared with White patients,¹³⁸ and studies in American

Indians, Alaskan Natives, and Hispanic Americans show disparities in risk for diagnosis with metastatic disease, and survival differences that were partially accounted for by neighborhood factors.¹³⁹ It is also possible that the COVID-19 pandemic further accelerated the disparities.¹⁴⁰ Ongoing studies are exploring the patterns of access to treatments and the financial burden imposed by the cost of care for treatments for mRCC. Continued attention to demographic and socioeconomic forces that influence care access is essential.

MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient may have two or more such conditions—referred to as multiple chronic conditions (MCC)—is challenging. Patients with MCC are a complex and heterogeneous population, making it difficult to account for all of the possible permutations to develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude these patients to avoid potential interaction effects or confounding of results associated with MCC. As a result, the reliability of outcome data from these studies may be limited, thereby creating constraints for expert groups to make recommendations for care in this heterogeneous patient population.

Since many patients for whom guideline recommendations apply present with MCC, any treatment plan needs to take into account the complexity and uncertainty created by the presence of MCC, highlighting the importance of shared decision making regarding guideline use and implementation. Therefore, in consideration of recommended care for the target index condition, clinicians should review all other chronic conditions present in the patient and take those conditions into account when formulating the treatment and follow-up plan.

In light of these considerations, practice guidelines should provide information on how to apply the recommendations for patients with MCC, perhaps as a qualifying statement for recommended care. This may mean that some or all of the recommended care options are modified or not applied, as determined by best practice in consideration of any MCC.

COST IMPLICATIONS

Having health insurance can affect the stage at which ccRCC is diagnosed and therefore the treatments needed.¹⁴¹ However, individuals with cancer are required to pay a larger proportion of their treatment costs through deductibles and coinsurance, and the costs of RCC treatment have been increasing over time.¹⁴²⁻¹⁴⁴ Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer treatments.^{145,146} Financial toxicity remains an understudied yet significant problem for patients with metastatic RCC.¹⁴⁷

Discussion of cost can be an important part of shared decision making.¹⁴⁸ Clinicians should discuss with patients the use of less expensive alternatives when it is practical and feasible for treatment of the patient's disease and there are two or more treatment options that are comparable in terms of benefits and harms.¹⁴⁸

Patient out-of-pocket costs may vary depending on insurance coverage. Coverage may originate in the medical or pharmacy benefit, which may have different cost-sharing arrangements. Patients should be aware that different products may be preferred or covered by their particular insurance plan. Even with the same insurance plan, the price may vary between different pharmacies. When discussing financial issues and concerns, patients should be made aware of any financial counseling services available to address this complex and heterogeneous landscape.¹⁴⁸

As part of the guideline development process, ASCO may opt to search the literature for published cost-effectiveness analyses that might inform the relative value of available treatment options. Excluded from consideration are cost-effective analyses that lack contemporary cost data; agents that are not currently available in either the United States or Canada; and/or are industry-sponsored. The following cost-effectiveness analysis was identified to inform the topic.^{149,150}

Comparisons of Nivo plus Ipi versus sunitinib and Nivo plus Ipi versus axitinib plus pembrolizumab versus sunitinib both found Nivo plus Ipi to be the most cost-effective combination in treating mRCC.^{151,152} However, in the absence of widely agreed upon definitions for incremental cost-effectiveness ratios and willingness to pay thresholds per quality-adjusted life-years gained, these analyses are limited. The number of potential treatments currently available and the nuanced clinical judgment used by providers to recommend one over the others also make comprehensive comparisons challenging. Despite these difficulties, discussions of treatment expense, especially out-of-pocket costs, in the context of overall financial toxicity (time off from work, AE management, travel, parking, childcare, elder care needs, etc) with the assistance of financial counselors and social workers can help with shared decision making and bolster informed consent.

EXTERNAL REVIEW AND OPEN COMMENT

The draft recommendations were released to the public for open comment from January 18, 2022, through February 2, 2022. Response categories of "Agree as written," "Agree with suggested modifications" and "Disagree. See comments" were captured for every proposed recommendation with 59 written comments received on the 20 recommendations from 23 respondents. Respondents either agreed or agreed with slight modifications to 75% of the recommendations and some respondents reported disagreements with 25% of the recommendations. Expert

Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions. After Expert Panel review of the Open Comment results, five recommendations were revised. All changes were incorporated before EBMC review and approval.

The draft was submitted to two external reviewers with content expertise. It was rated as high quality, and it was agreed it would be useful in practice. Review comments were reviewed by the Expert Panel and integrated into the final manuscript before approval by the EBMC.

GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a member from ASCO's Practice Guideline Implementation Network (PGIN) on the panel (Dr Rubina Qamar was the PGIN representative on this guideline panel). The additional role of this PGIN representative on the guideline panel is to assess the suitability of the recommendations for implementation in the community setting (see the Data Supplement 8: Practice Guideline Implementability Network Review Results), but also to identify any other barriers to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among front-line practitioners and survivors of cancer and caregivers, and also to provide adequate services in the face of limited resources. No barriers to implementation of the guideline recommendations were reported in the PGIN survey for this guideline.

The guideline Bottom Line Box was designed to facilitate implementation of recommendations. This guideline will be distributed widely through the ASCO PGIN. ASCO guidelines are posted on the ASCO website and are most often published in the *Journal of Clinical Oncology*.

LIMITATION OF THE RESEARCH AND FUTURE RESEARCH

There are numerous limitations to the application of a broad and comprehensive set of guidelines from a collection of discrete studies, aiming to achieve consensus from a diverse panel of individual stakeholders and experts. This guideline aims to provide the best guidance for practitioners at this point in time, and a framework that will facilitate clinical decision making from now until these guidelines are updated. The following is a segment of factors that are limiting in interpreting these guidelines.

First and foremost, these guidelines apply exclusively to patients with ccRCC. The unfortunate reality is that although clear cell histology only applies to roughly 70% of cases, the majority of clinical trials are restricted to this disease subset. Therefore, extrapolating to non-ccRCC is not possible, and as we learn more about the biology of

these diverse malignancies, we know that such extrapolation is often not even logical. The diagnosis of renal cell carcinoma not otherwise specified is a particularly challenging issue that must be reconciled with more precise diagnostics.

A second major limitation is the lack of a universal standard for comparison. The field has moved rapidly, and as a result, the majority of trials discussed in this guideline were compared with sunitinib as the standard of care for first line. The standard evolved and has created an imperfect comparator. These guidelines are therefore reviewed with a lens that has the advantage of being able to know if populations had similar responses to sunitinib, but without having effective comparisons to modern-day treatments and outcomes.

Perhaps, the most significant barrier to the application of these guidelines is the degree to which these treatments were studied in populations that represent the community of patients with metastatic ccRCC at large. Clinical trial participation hovers around 3% of eligible participants and is limited to selected centers. Even with the expansion of larger trials to community practice sites, the demographics of clinical trial participants is more often insured, more likely to live in urban or suburban centers, more White, and younger than the overall population.¹⁵³ These important distinctions must be considered in applying treatment decisions outside clinical trials. Specific patient groups, such as those with brain metastases, are variably excluded from studies, but these patients are regularly encountered in real life. In the current immunotherapy era, a study published in 2021 which was highly represented by renal cell carcinoma patients, found approximately 32% of patients failed to meet standard eligibility criteria.¹⁵⁴ Analyses of these populations suggest that overall, responses are inferior to the reported outcomes.¹⁵⁵ Thus, generalization of findings to the population at large raises several challenging limitations, and must be applied with caution.

Finally, how we define patients on the basis of risk for recurrence, response, or death from renal cell cancer relies upon criteria that were created in the era of treatment solely with TKIs¹⁵⁶ or earlier (ie, MSKCC risk criteria).¹⁵⁷ These models are in widespread use, and create a foundation for our communication about patients on and off of clinical protocols. A revised model that considers response to immunotherapy, particularly in the era of double immunotherapy, is needed.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

ADDITIONAL RESOURCES

More information, including a supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/genitourinary-cancer-guidelines. Patient information is available at www.cancer.net.

RELATED ASCO GUIDELINES

- Integration of Palliative Care into Standard Oncology Care¹⁵⁸ (<http://ascopubs.org/doi/10.1200/JCO.2016.70.1474>)
- Patient-Clinician Communication¹³² (<http://ascopubs.org/doi/10.1200/JCO.2017.75.2311>)
- Management of Small Renal Masses¹⁵⁹ (<https://ascopubs.org/doi/full/10.1200/JCO.2016.69.9645>)

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EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, slide sets, clinical tools and resources, and links to patient information at www.cancer.net, is available at www.asco.org/genitourinary-cancer-guidelines.

EQUAL CONTRIBUTION

W.K.R. and P.J.V.V. were Expert Panel coauthors. E.A.S. served as coauthor from March 30, 2020, through March 22, 2022.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Conception and design: All authors
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Provision of study materials or patients: All authors
Collection and assembly of data: All authors
Data analysis and interpretation: All authors
Manuscript writing: All authors
Final approval of manuscript: All authors
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 Expert Panel members are listed in Appendix [Table A2](#) (online only).

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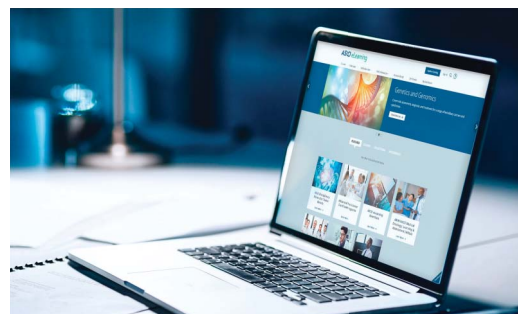
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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline**

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Research Funding: Astellas Medivation

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APPENDIX

TABLE A1. Recommendation Rating Definitions

Term	Definitions
Quality of evidence	
High	We are very confident that the true effect lies close to that of the estimate of the effect
Moderate	We are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect
Very low	We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect
Strength of recommendation	
Strong	In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects
	In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects
	All or almost all informed people would make the recommended choice for or against an intervention
Weak	In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists
	In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists
	Most informed people would choose the recommended course of action, but a substantial number would not

TABLE A2. Management of Metastatic Clear Cell Renal Cell Carcinoma: ASCO Guideline Expert Panel Membership

Name	Institution	Specialty
W. Kimryn Rathmell, MD, PhD, Co-chair (March 30, 2020 to present)	Vanderbilt University Medical Center, Nashville, TN	Medical oncology, urology
Peter J. Van Veldhuizen, MD, Co-chair (March 22, 2022 to present)	University of Rochester Medical Center, Rochester, NY	Medical oncology
Eric A. Singer, MD, MA, MS, immediate past Co-chair (March 30, 2020, through March 22, 2022)	Rutgers Cancer Institute of New Jersey, New Brunswick, NJ	Urologic oncology
Hikmat Al-Ahmadie, MD	Memorial Sloan Kettering Cancer Center, New York, NY	Pathology
Hamid Enamekhoo, MD	University of Wisconsin Carbone Cancer Center, Madison, WI	Medical oncology
Ralph J. Hauke, MD	Nebraska Cancer Specialists, Omaha, NE	Hospice and palliative medicine, medical oncology
Terry M. Kungel, MBA	Woolwich, ME	Patient representative
Bryan Lewis, JD	KidneyCan, Philadelphia, PA	Patient representative
Alexander V. Louie, MD, MSc	Odette Cancer Centre, Sunnybrook Health Sciences Centre, Toronto, ON, Canada	Radiation oncology
Matthew I. Milowsky, MD	University of North Carolina at Chapel Hill, Chapel Hill, NC	Medical oncology
Ana M. Molina, MD	Weill Cornell Medicine, New York, NY	Medical oncology
Rubina Qamar, MD	Advocate Aurora Health, Milwaukee, WI	Medical oncology, practice guidelines implementation network representative
Tracy L. Rose, MD, MPH	University of North Carolina at Chapel Hill, Chapel Hill, NC	Medical oncology
Shankar Siva, MD, PhD	Peter MacCallum Cancer Center, Melbourne, Australia	Radiation oncology
Nicholas G. Zaorsky, MD, MS	Department of Radiation Oncology, University Hospitals Seidman Cancer Center, Case Western Reserve School of Medicine, Cleveland, OH	Radiation oncology
Tian Zhang, MD	Harold C. Simmons Comprehensive Cancer Center, UT Southwestern Medical Center, Dallas, TX	Medical oncology
R. Bryan Rumble, MSc	American Society of Clinical Oncology, Alexandria, VA	ASCO practice guideline staff (health research methods)